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MILITARY SPECIFICATION

STEERING CONTROL SYSTEMS, NAVAL SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements applicable in designing and constructing shipboard steering control systems intended for use with electrohydraulic steering gear equipment in accordance with MIL-S-17803. This specification defines performance criteria, the environmental conditions within which the equipment must operate satisfactorily and reliably, the process for selection and application of general material and parts, and the means by which equipment as a whole will be tested to determine if it is acceptable to the Navy. Reference to a ship steering control system herein shall mean the complete unit, including control equipment located in the steering gear room and on the bridge. This specification covers only one classification of steering control system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

TT-C-490	Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings
TT-P-664	Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, VOC-Compliant
TT-P-1757	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity

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MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-T-713	Twine, Fibrous: Impregnated, Lacing and Tying
MIL-S-901	Shock Tests, HI (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-12285	Switches, Thermostatic
MIL-P-15024	Plates, Tags and Bands for Identification of Equipment
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-A-15303	Audible Signals: Alarms, Bells, Buzzers, Horns, and Sirens, Electronic, Shipboard
DOD-P-15328	Primer (Wash), Pretreatment (Formula No. 117 for Metals); (Metric)
MIL-S-15743	Switches, Rotary, Enclosed
MIL-F-16552	Filter, Air Environmental Control System, Cleanable, Impingement (High Velocity Type)
MIL-I-16923	Insulating Compound, Electrical, Embedding
MIL-C-17361	Circuit Breakers, Air, Electric, Insulated Housing (Shipboard Use), General Specification for
MIL-C-17588	Circuit Breakers (Automatic - ALB-1) and Switch, Toggle (Circuit Breaker, Non-Automatic - NLB-1) Air, Insulated Housing, 125 Volts and Below, AC and DC, (Naval Shipboard Use)

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MIL-S-17803	Steering Gears, Electro-Hydraulic, Marine
MIL-S-18396	Switches, Meter and Control, Naval Shipboard
MIL-S-20708	Synchros, General Specification for
MIL-S-21604	Switches, Rotary, Multipole and Selector; General Specification for
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for
MIL-T-23648	Thermistor, (Thermally Sensitive Resistor) Insulated, General Specification for
MIL-R-28750	Relays, Solid State, General Specification for
MIL-M-28787	Modules, Standard Electronic, General Specification for
MIL-C-39018	Capacitors, Fixed, Electrolytic (Aluminum Oxide), Established Reliability and Non-Established Reliability, General Specification for
MIL-R-39023	Resistors, Variable, Non-Wire-Wound, Precision General Specification for
MIL-T-43435	Tape, Lacing and Tying
MIL-S-83731	Switches, Toggle, Unsealed and Sealed Toggle, General Specification for

STANDARDS

MILITARY

MIL-STD-12	Abbreviations for Use on Drawings, and in Specifications, Standards and Technical Documents
MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-198	Capacitors, Selection and Use of
MIL-STD-199	Resistors, Selection and Use of
MIL-STD-220	Method of Insertion-Loss Measurement
MIL-STD-242	Electronic Equipment Parts Selected Standards
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

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MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-483	Configuration Management Practices for Systems, Equipment, Munitions, and Computer Programs
MIL-STD-490	Specification Practices
MIL-STD-721	Definitions of Terms for Reliability and Maintainability
MIL-STD-740-1	Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-740-2	Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-1132	Switches and Associated Hardware, Selection and Use of
MIL-STD-1251	Screws and Bolts Preferred for Design, Listing of
MIL-STD-1310	Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety
MIL-STD-1346	Relays Selection and Application
MIL-STD-1353	Electrical Connectors, Plug-In Sockets and Associated Hardware, Selection and Use of
MIL-STD-1378	Requirements for Employing Standard Electronic Modules
MIL-STD-1389	Design Requirements for Standard Electronic Modules
MIL-STD-1399, Section 300	Interface Standard for Shipboard Systems Section 300 Electric Power, Alternating Current; (Metric)
DOD-STD-2167	Defense System Software Development

HANDBOOK

MILITARY

MIL-HDBK-225	Synchros Description and Operation

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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NORTH ATLANTIC TREATY ORGANIZATION (NATO)

STANAG 4156	Standard Specification for a Serial Data Interface for Synchronous Connections to a Data Network

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVMAT-P-9492	Navy Manufacturing Screening Program
S9407-AB-HDBK-010	Shipboard Electromagnetic Shielding Practices

(Application for copies should be addressed to the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI 753.1	Safety Color Code For Marking Physical Hazards

AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM)

ASTM F-1166	Standard Practice For Human Engineering Design For Marine Systems, Equipment, and Facilities

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

HIR 1063	General Environmental Considerations for Marine Vehicles; (DOD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. In the event of a conflict between the text of this specification and MIL-S-17803, MIL-S-17803 shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 General description and characteristics.

3.2.1 Purpose. The purpose of the steering control subsystem is to ultimately control the position of the ship rudders. The steering control subsystem converts steering commands to electrical signals which control the fluid to the hydraulic actuators. The hydraulic actuators position the rudders through mechanical linkages.

3.2.2 Major units. The steering control system shall contain the following units: (a) Bridge Steering Unit, (b) Aft Steering Unit, and (c) Local Control units.

3.2.2.1 Bridge Steering Unit (BSU). The BSU shall contain the equipment necessary to initiate and sustain complete control of any available mode of steering with maximum performance capability. Equipment shall be mounted in the ship control console in an arrangement which is compatible with other types of equipment mounted in the console. This unit shall contain a Steering Power Control Panel, Steering Mode Control Panel, helm angle indicator, Automatic Steering Control Panel, Emergency Override to Manual Panel, a manual steering wheel assembly, and interface electronics unit containing power supplies and circuitry to interface the elements of the Bridge Steering Unit to other elements of the steering control subsystem and to the dimming circuits in the control console.

3.2.2.2 Aft Steering Unit (ASU). The ASU shall be located in the steering gear room. It shall contain the equipment necessary to initiate and sustain complete control of any available mode of steering with maximum performance capability. This unit shall contain a Steering Power Control Panel, Steering Mode Control Panel, helm angle indicator, Automatic Steering Control Panel, Emergency Override to Manual Panel, and a manual steering wheel assembly. This unit shall also contain a digital controller, interface electronics, supporting dc power supplies, and a maintenance panel. Panels and assemblies shall be mounted in a console, and panel layout shall be similar to the ship control console on the bridge.

3.2.2.3 Local control units (LCUs). The LCUs shall be located in the steering gear room and shall contain the circuitry and switching necessary to interface ordered rudder angle signals with the hydraulic power units. Each unit shall provide the means for selecting the source of the ordered rudder angle signal used with its associated hydraulic power unit. There shall be one local control unit dedicated to each hydraulic power unit.

3.2.3 Performance.

3.2.3.1 Survivability features.

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3.2.3.1.1 Multiple steering stations. The control subsystem shall provide full performance steering of the ship from the following locations:

a. Bridge
b. Steering gear room.

3.2.3.1.2 Single casualty criteria. The steering control subsystem shall be designed and constructed so that if a single casualty causes a loss of rudder control, changing the power units by manipulating switches at the Bridge Steering Unit or the Aft Steering Unit shall cause full rudder control to be regained within 4 seconds after initiating the power unit transfer.

3.2.3.1.3 Short and open protection on control lines. The control subsystem shall be designed to ensure that a short to signal return or ships ground or open circuit along any control line will not result in damage to equipment or degradation of system performance.

3.2.3.1.4 Automatic configuration change. Automatic configuration change shall occur when a condition is detected that impairs operation in the selected mode. When a fault occurs in the Automatic mode, the system shall switch automatically to either the Manual mode (if available) or to the Backup Manual mode (if Manual mode is not available). If a fault occurs in the Manual mode, the system shall switch automatically to the Backup Manual mode. Switching to the Automatic mode shall be by manual means only.

3.2.3.2 Subsystem functional characteristics.

3.2.3.2.1 Control modes. The steering control subsystem shall have three modes of controlling the rudders as follows:

a. *BACKUP MANUAL.* This control mode shall permit the helmsman to manually initiate rudder commands by means of a hand steering wheel. Operation of the control system in this mode shall be completely independent of the digital controller. This control mode shall be considered the lowest remote control configuration.

b. *MANUAL.* This control mode shall permit the helmsman to manually enter rudder commands by means of a hand steering wheel in the same manner as in the Backup Manual mode. The digital controller shall monitor these manually generated rudder angle commands and generate a composite rudder angle command based upon steering control algorithms. This control mode shall be considered a higher configuration than the Backup Manual mode, but a lower configuration than the Automatic mode.

c. *AUTOMATIC.* This control mode shall permit the helmsman to enter heading commands, and the control system shall respond by generating appropriate rudder angle commands to maintain the ordered heading. The control system shall utilize a digital controller to assemble data from the ship navigation equipment along with the ordered

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heading entered by the helmsman, and compute the rudder angle commands necessary to cause the ship to maintain the ordered heading. This control mode shall be considered the highest remote control configuration.

3.2.3.2.2 Control of hydraulic components.

3.2.3.2.2.1 Electric motors. Each hydraulic power unit shall have a main pump electric motor, as required by MIL-S-17803. The control of this electric motor shall incorporate the following features:

a. A means shall be provided to remotely activate and deactivate the electric motor in each hydraulic power unit.

b. The actual ON-OFF status of each electric motor shall have a positive indication on the bridge and in the steering gear room.

3.2.3.2.2.2 Control of transfer valves. As required by MIL-S-17803, each hydraulic power unit shall have a remotely operated transfer valve. Control of this transfer valve shall have the following features:

a. *Remote Control.* A means shall be provided to remotely activate and deactivate the transfer valves in each hydraulic power unit.

b. *Status Presentations.* A means shall be provided on the bridge and in the steering gear room to indicate which hydraulic power unit is engaged.

3.2.3.2.2.3 Servo valve control. The servo valve in each hydraulic power unit shall have dual coils. A means shall be provided to energize these coils with currents generated by a dual set of electronics located within each local control unit. Only one coil shall be energized at any one time.

3.2.3.2.3 Control transfer features. The transfer of control from one steering unit to another shall be made in such a manner that the ship steering shall not be adversely modified due to the transfer. When operating in the Backup Manual mode, coordinated transfer between steering stations shall not be possible. Control transfer shall incorporate the following features:

a. *Normal operation.* Normal operation shall be defined as operation with all steering units and interconnecting mediums intact and operating with full performance. Emergency situations are not considered normal operations as defined herein. During normal operation, coordinated transfer of control shall be possible only between the Bridge Steering Unit and the Aft Steering Unit. When transfer of control is requested, the STATION CONTROL (BRIDGE) or STATION CONTROL (AFT) indicators at both steering stations shall flash, indicating which station is requesting control. Indicators at both steering stations shall illuminate continuously, indicating which station is controlling the steering. During normal operation, all status indicators shall be active regardless of which unit has control of the steering.

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b. *Emergency operations.* Emergency operations shall be defined as operations required as a result of casualty, equipment failure or malfunction. An EMERGENCY OVERRIDE TO MANUAL switch shall be provided on the bridge and in the steering gear room, which when pressed, will cause an immediate transfer of control to the unit where the switch is pressed. The control mode following the emergency transfer shall be Backup Manual.

c. *Coordinated control transfer scenario.* For the purpose of this description, it shall be assumed that the Bridge Steering Unit has control and the Aft Steering Unit wants control of the steering. The STATION CONTROL (AFT) push button at the Aft Steering Unit shall be pressed and held to request control. The STATION CONTROL (AFT) indicators at all steering units will flash, indicating that the Aft Steering Unit is requesting control. Since the Bridge has control, concurrent depression of the STATION CONTROL (AFT) push button in the Bridge Steering Unit will effect an immediate transfer of steering control to the Aft Steering Unit. When a control transfer occurs in the Automatic mode, the heading order transferred shall be the ordered ship heading at the time of transfer.

3.2.3.2.4 Fault classifications. The digital controller shall be capable of detecting system malfunctions and classifying them as critical or noncritical system faults.

3.2.3.2.4.1 Critical system faults. A critical system fault shall be defined as a fault resulting from any malfunction that contributes to the generation of an erroneous rudder angle order command.

3.2.3.2.4.2 Noncritical system fault. Noncritical system faults shall be classified as either major or minor according to the following criteria:

a. *Major* – A noncritical failure which results in loss of at least one steering control mode

b. *Minor* – A noncritical, nonmajor condition requiring attention.

3.2.3.2.4.3 Critical hydraulic power unit faults. A critical hydraulic power unit fault shall be defined as a fault resulting from any malfunction that causes an automatic shutdown of the operating unit and automatic startup of the alternate unit.

3.2.3.2.5 Fault displays. Critical fault and major fault displays shall be both visual and audible. Minor fault displays shall be only visual. When a major fault or critical fault occurs, the relevant visual indicator at the Bridge and Aft Steering Units shall flash and the audible sound shall be continuously emitted at these steering stations. The alarm shall be acknowledged by pressing the flashing indicator identifying the alarm. Pressing the flashing indicator shall silence the audible alarm at the station where the switch-indicator was pressed. The indicator at both stations shall continue to flash until the fault is acknowledged at the other station. When the indicator is pressed at the other station, the audible alarm shall stop and the relevant indicators at both stations shall display constant illumination until the fault is cleared. When more than one critical or major fault occurs before any fault acknowledgement, the audible alarm shall be silenced with the first fault acknowledgement. Critical

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or major faults occurring after the acknowledgement shall retrigger the audible alarm. When a minor fault occurs, the relevant visual indicator at the steering stations shall flash. The fault shall be acknowledged by pressing the flashing fault indicator. The indicator shall continue to flash at the steering stations until the fault is acknowledged at both stations. When the fault is acknowledged at both stations, the relevant indicators shall display constant illumination until the fault is cleared. The control system shall provide, as a minimum, the following fault displays:

a. *Digital subsystem.* The occurrence of any one or more of the following conditions shall activate both a visual and audible alarm at all steering stations:

(1) Digital controller fault

(2) Faulty acquisition of the ship information, regardless of the cause

(3) Automatic steering malfunction. Any condition that results in the failure to maintain the ship heading within limits specified herein

(4) Failure to accurately transfer data over digital data transmission paths.

b. *Hydraulic power unit (critical faults).* The occurrence of one or more of the following conditions in any of the hydraulic power units shall be considered a power unit critical fault, and shall activate a visual and audible indicator on the bridge and in the steering gear room and also cause an automatic transfer to the alternate power unit:

(1) Low low oil level in head tank
(2) High high temperature in main pump case drain piping
(3) Electric motor overload
(4) Loss of alternating-current (ac) power.

c. *Servo valve control fault.* When two machinery subsystems are required, the magnitude of the current in each of the energized servo valve coils shall be monitored by the digital controller and if a significant difference is detected, a visual fault indicator shall be activated on the bridge and in the steering gear room.

3.2.3.2.6 Audible alarms. A continuous audible alarm shall be emitted at all steering stations when a critical fault or major fault occurs within the control subsystem. Audible alarms shall operate in accordance with 3.2.3.2.5.

3.2.3.2.7 Maintenance modes. The control system shall be considered in a maintenance mode when any type of maintenance activity (preventive or corrective) is in progress on any unit within the control system. A scheduled maintenance operation within the control system shall not cause a loss of all steering modes. At least one steering mode shall be available at all times. Maintenance operations on the control system shall have the following characteristics:

a. Maintenance actions performed on the digital controller or associated data transmission system shall not affect system operation when the system is in the Backup Manual mode.

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b. Maintenance actions taken on a local control unit or its associated hydraulic power unit shall not cause a loss of operational performance in the system.

c. Maintenance on the control system shall consist of isolating the fault to an electronic module assembly, replacing that assembly, and verifying that proper operations have been restored.

d. *Status presentations.* Any action which places switches at a local control unit or motor controller in any position other than NORMAL shall be considered a maintenance action on the associated hydraulic power unit. The associated NOT READY indicators shall illuminate at the Bridge and Aft Steering Units to alert the operator that the power unit is not available.

e. *Maintenance panel.* A maintenance panel, which shall be located in the steering gear room, shall consist of the necessary indicators and displays to provide maintenance personnel with information related to normal system performance, as well as information to expedite fault location.

f. *Built-in test.* The digital controller within the steering gear room console shall monitor the entire system and display this fault information to the maintenance personnel in a self-explanatory manner so that neither code conversions nor reference to any type of look-up table shall be required.

3.2.3.2.8 Data transfer features. The exchange of data between steering units, or between steering units and ship information sources, shall be made at such a rate that maximum system performance is not impaired under any ship handling situations.

3.2.3.2.8.1 Data update rates. The control subsystem shall be capable of transmitting data at the rates necessary to meet performance requirements.

3.2.3.2.8.2 Transmission. The steering control subsystem shall incorporate two methods of transferring data. One method utilizes hard wire cabling and the other utilizes digital data transmission. When compatibility with the Data Multiplex System (DMS) is required (see 6.2), digital data transmission shall utilize DMS. Transmission of rudder angle commands from the potentiometers in the steering wheel assembly shall be hard wired between remote units and the steering gear room. Transmission of these signals shall be divided between port and starboard cables in such a manner as to ensure maximum survivability.

3.2.3.2.8.3 Inter-unit communication. Inter-unit communications paths shall be provided between the major units of the Steering Control subsystem and the ship information sources. Discrete signals shall be provided between the Bridge Steering Unit and each local control unit (LCU) which are independent of the Aft Steering Unit. Ship information sources necessary to perform steering control functions and appropriate Bridge Steering Unit functions shall be interfaced with the digital

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controller. Redundant communication paths from the digital controller to both the Bridge Steering Unit and to the ship information sources shall be provided. A single communication path shall be provided between each of the following:

a. The digital controller and each LCU
b. The digital controller and the After Interface Electronics Unit (AIEU).

The digital controller shall attempt to recover from inter-unit communication errors by retransmitting messages over the original path. If the original path fails, re-transmission over an alternate path, if available, shall be attempted. Inter-unit communication errors shall be monitored in order to identify and isolate hardware faults. When a fault is found, the digital controller shall automatically determine when the fault has been corrected, and then automatically resume normal operation. When both inter-unit communication paths from the digital controller to the Bridge Steering Unit, or from the digital controller to the ship information sources are working, the digital controller shall send and receive messages using both paths, and messages shall be assigned to paths as necessary to equalize the traffic on both paths.

3.2.3.2.9 Indicator, display, and panels illumination. The requirements of this paragraph shall apply to:

a. Steering Power Control Panel (SPCP) (see 3.2.3.3.1)
b. Steering Mode Control Panel (SMCP) (see 3.2.3.3.2)
c. Automatic Steering Control Panel (ASCP) (see 3.2.3.3.3)
d. Emergency Override to Manual Panel (EOTMP) (see 3.2.3.3.5).

3.2.3.2.9.1 Indicators. Indicators on the SPCP, SMCP, and ASCP shall be red when illuminated. For those which are to have always-visible legends, the legends shall be reflective white. Legends shall be as large as possible; abbreviations shall be in accordance with MIL-STD-12.

Indicators shall fall into four categories, depending upon their functions. Indicators in categories 1, 2, and 4 shall not be capable of being dimmed to extinction. The categories are as follows:

a. *Category 1.* Indicators/switches, normally dead-faced, which illuminate and then require operation. No other night-time guidance shall be necessary.

(1) All fault indicators ..... SPCP, SMCP
(2) STATION CONTROL (also falls in category 2) ..... SMCP

b. *Category 2.* Indicator/switches, normally dead-faced, which are manually operated to set a new state of system operation, and then remain illuminated until another state is selected. These shall require always-visible border lines on the panels to guide the operator to them.

(1) STATION CONTROL (also falls in category 1) ..... SMCP
(2) MODE CONTROL ..... SMCP

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(3) CONTROL ..... SPCP
(4) LEFT TURN, RIGHT TURN ..... ASCP
(5) PERFORMANCE ..... ASCP
(6) HEADING MONITOR ..... SMCP

c. *Category 3.* Indicator/switches which are always visible (white reflective - day/red light - night). Lighting for these indicators shall be dimmed with the panel, not with the other indicators. The indicators on these switches are not related to a state and shall be illuminated only at night to guide the operator to them. No other night-time guidance needed shall be necessary.

(1) Keyboard ..... ASCP
(2) LEFT (EXECUTE), RIGHT (EXECUTE) ..... ASCP
(3) ALARM TEST ..... SMCP
(4) LAMP TEST ..... SMCP
(5) RUDDER LIMIT ..... ASCP

d. *Category 4.* Indicators only, normally dead-faced.

(1) READY (AUTO) ..... SMCP
(2) READY (MANUAL) ..... SMCP
(3) READY (STATUS) ..... SPCP
(4) NOT READY (STATUS) ..... SPCP

3.2.3.2.9.2 Displays. Numeric displays, if of the illuminated-segment type (for example, light emitting diode (LED)), shall be dimmed with category 1, 2, and 4 indicators and shall not be capable of being dimmed to extinction. If they are of the transflexive type (for example, liquid crystal display (LCD)), their back lighting shall be dimmed with the panels.

3.2.3.2.9.3 Panels. Panels shall be in accordance with MIL-P-7788, class 1-R, type V. The panels shall have illuminated lines on at least three sides of each category 2 indicator as a means of guiding the operator's finger to the associated switch actuator. Abbreviations shall be avoided, but if used shall be in accordance with MIL-STD-12. With the exception of the keyboard on the ASCP, there shall be a panel legend associated with each indicator or switch/indicator. These legends may be shared by more than one indicator if they are similar in function (for example, the READY indicators on the SPCP).

3.2.3.2.10 Automatic steering.

3.2.3.2.10.1 General. Automatic steering (autopilot) shall provide optimum heading-keeping and maneuvering performance. Automatic steering shall be self-adaptive and shall require no operator adjustments over the complete range of ship operating conditions. It shall maintain the ordered ship

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heading in any sea state up to sea state 3, with a speed of 2/3 ahead or greater into a head sea. The ship heading shall not overshoot the ordered heading by more than one degree for any ordered heading change. Steering control algorithms shall optimize rudder movement for minimum fuel consumption and heading changes at all ship speeds and rudder angle limits.

3.2.3.2.10.2 Stability. The steering control system shall be demonstrated by analysis and testing to be stable in all modes of operation and in all specified environmental conditions. The stability margin shall include a gain margin of not less than 10 decibels (dB) and a phase margin of not less than 45 degrees. There shall be no backlash, deadzone oscillation, or limit cycle oscillation.

3.2.3.2.10.3 Maneuvering. Automatic steering shall accomplish heading changes in minimum time with not more than one degree overshoot for any given rudder angle limit value and at all ahead ship speeds.

3.2.3.2.10.4 Heading keeping. Automatic steering shall minimize the root-mean-square (rms) heading deviations for any given ordered heading. When the system is operating in the lowest gain (BEST FUEL) performance mode, the ship fuel economy shall be the maximum attainable. When operating in the highest gain (BEST HEADING) performance mode, the ship rms heading deviations shall be held to a minimum. Automatic steering shall also provide an INTERMEDIATE performance mode which allows the ship to operate between the lowest and highest gain values of the system.

3.2.3.2.10.5 Adaptability. Software which implements the automatic steering functions shall be modularized so that parameters which are particular to the ship class can be changed without the need to alter other modules.

3.2.3.2.10.6 Simulation. Engineering estimates regarding automatic control system stability and performance shall be validated by means of a dynamic computer simulation. Significant dynamic effects and nonlinearities in the control and machinery subsystems shall be taken into account. The simulation shall demonstrate that the self-adaptive features provide optimum maneuvering, heading keeping, and fuel saving performance over the complete range of ship operating conditions. When algorithms for roll stabilization are required (see 6.2), the simulation shall also demonstrate that the self-adaptive features minimize rudder movements that would induce or contribute to ship roll.

The dynamic analysis shall demonstrate the operation of the steering control system, identify initial steering control set points, and indicate how the analysis was used in the design process. Simulation mathematical models used in the dynamic analysis shall include the following elements as a minimum:

a. Ship

(1) Displacement
(2) Total resistance versus ship speed
(3) Trimmed −5 to +5 degrees
(4) List up to 15 degrees

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(5) Rudder characteristics and limits
(6) Up to 10 degrees pitch
(7) Up to 45 degrees roll.

b. Control system

(1) Optimum maneuvering performance
(2) Optimum heading keeping performance
(3) Intermediate performance
(4) Self-adaptive features for ship speed, sea state, and wind
(5) Mode transitions
(6) Rudder limits
(7) Hydraulic system characteristics.

c. Propulsion system configuration evaluated as part of the dynamic analysis shall include:

(1) Two screw operation (if appropriate)
(2) Single-screw operation (opposite shaft trailing).

d. Ship maneuvers evaluated as part of the dynamic analysis shall include the following:

(1) Steady-state minimum to maximum ahead power for each propulsion system configuration

(2) Maximum to minimum ahead for each propulsion system configuration

(3) Automatic turn execution at minimum and maximum rate to new courses 10 through 360 degrees from the original course.

e. Environmental conditions evaluated as part of the dynamic analysis shall include sea states up to 3 and equivalent wind speeds.

3.2.3.2.10.6.1 Dynamic analysis reports. Dynamic analysis reports shall identify as early as possible the mathematical models to be used, the analytical approach, conditions to be simulated, interim results and associated control system design additions/deletions/modifications recommended or accomplished based on simulation results. The final dynamic analysis report shall include the final mathematical models, analytical approach, conditions simulated and results.

3.2.3.2.10.6.2 Final analysis. The final dynamic analysis shall include simulation runs conducted using the final control system configuration with recommended system settings. These simulation runs shall be representative of all the types of runs made during the design development and shall demonstrate proper control system/steering system/ship response for all analyzed conditions. The following information shall be evaluated:

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a. Control system schedules, gains, setpoints, and sequencing logic
b. Steady-state system performance
c. Ship maneuvering performance
d. Fuel optimization performance.

3.2.3.2.10.7 Prohibited operation. The automatic steering control shall be designed so that operation in the Automatic mode is prohibited at ship speeds of 2 m/s (meters per second) (4 knots) or less. When operating in the Automatic mode and ship speed drops to 2 m/s (4 knots) or less, the system shall switch automatically to the Manual mode (if working) or to the Backup Manual mode. The MODE CONTROL (MANUAL) or MODE CONTROL (BACKUP MANUAL) indicator shall flash and the audible alarm shall sound in accordance with 3.2.3.3.2.4.a or 3.2.3.3.2.4.d.

3.2.3.3 Subsystem components.

3.2.3.3.1 Steering Power Control Panel.

3.2.3.3.1.1 Physical characteristics. The maximum panel height, including mounting flange, shall be not greater than 360 mm (millimeters) (14 inches). The maximum panel width, including mounting flange, shall be not greater than 230 mm (9 inches). The maximum depth behind the panel, including connectors, shall be not greater than 200 mm (8 inches).

3.2.3.3.1.2 Functional characteristics. The following functional characteristics shall be provided by the Steering Power Control Panel:

a. Shall remotely start and stop the electric motors in each hydraulic power unit.

b. Shall provide a positive visual indication of the on or off status of the electric motor in each hydraulic power unit.

c. Shall provide a positive visual indication of the electric power status in each hydraulic power unit.

d. Shall provide a positive visual indication of which hydraulic power unit is active (connected to the actuator).

e. Shall provide a positive visual and audible indication when any critical fault occurs in any hydraulic power unit. (See 3.2.3.2.4 for definition of critical faults in hydraulic power units.)

f. Shall provide means for acknowledging individual audible alarms as specified in 3.2.3.3.1.3.d.

g. This equipment shall be interchangeable with all other Steering Power Control Panels regardless of where they are mounted. Once mounted, the equipment shall operate properly for the steering station in which it is mounted without modification.

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h. When compatibility with DMS is required (see 6.2), all data communications between the SPCP in the BSU and the local control units shall utilize DMS switch closure discrete interfaces. Data communications between this panel and the digital controller shall pass through the interface electronics and shall utilize DMS STANAG 4156 (protocol type A) interfaces. The DMS STANAG 4156 (protocol type A) interfaces are used to provide redundant capability for the discrete interface. When compatibility with DMS is not required, all data communications between the SPCP in the BSU and the local control units shall be in accordance with 3.2.3.2.8.3.

3.2.3.3.1.3 Controls and switches. The following controls and switches shall be accessible at the Steering Power Control Panel (see figure 1 for typical panel layout).

a. *START MOTOR.* This switch shall be used to start the electric motor in the associated hydraulic power unit. This switch shall not be active unless the READY indicator is illuminated on the Steering Power Control Panel at the controlling station. There shall be a START MOTOR switch for each hydraulic power unit. Each switch shall be labeled to identify its corresponding hydraulic power unit.

b. *STOP MOTOR.* This switch shall be used to stop the electric motor in the associated hydraulic power unit, and start the motor and engage the rudder in the alternate hydraulic power unit. There shall be a STOP MOTOR switch for each hydraulic power unit. Each switch shall be labeled to identify its corresponding hydraulic power unit.

c. *ENGAGE RUDDER.* This switch shall be used to connect the associated hydraulic power unit to the actuator and disconnect the alternate power unit from the actuator. Depressing this push button shall also start the associated electric motor. There shall be an ENGAGE RUDDER switch for each hydraulic power unit. Each switch shall be labeled to identify its corresponding hydraulic power unit.

d. *FAULT ACKNOWLEDGE.* Switches shall be provided to acknowledge the visual alarm and mute the audible alarm resulting from a fault condition. These switches shall be combined with each fault indicator and, when pressed, shall mute the alarm associated with the particular fault indicator (see 3.2.3.2.5).

3.2.3.3.1.4 Indicators. The following indicators shall be mounted on the front panel of the Steering Power Control Panel. The brightness of each indicator shall be controlled by the master dimmer control serving the console in which the panel is located. Indicators associated with panels on the ASU Console need not be dimmed. (See figure 1 for a typical panel layout.)

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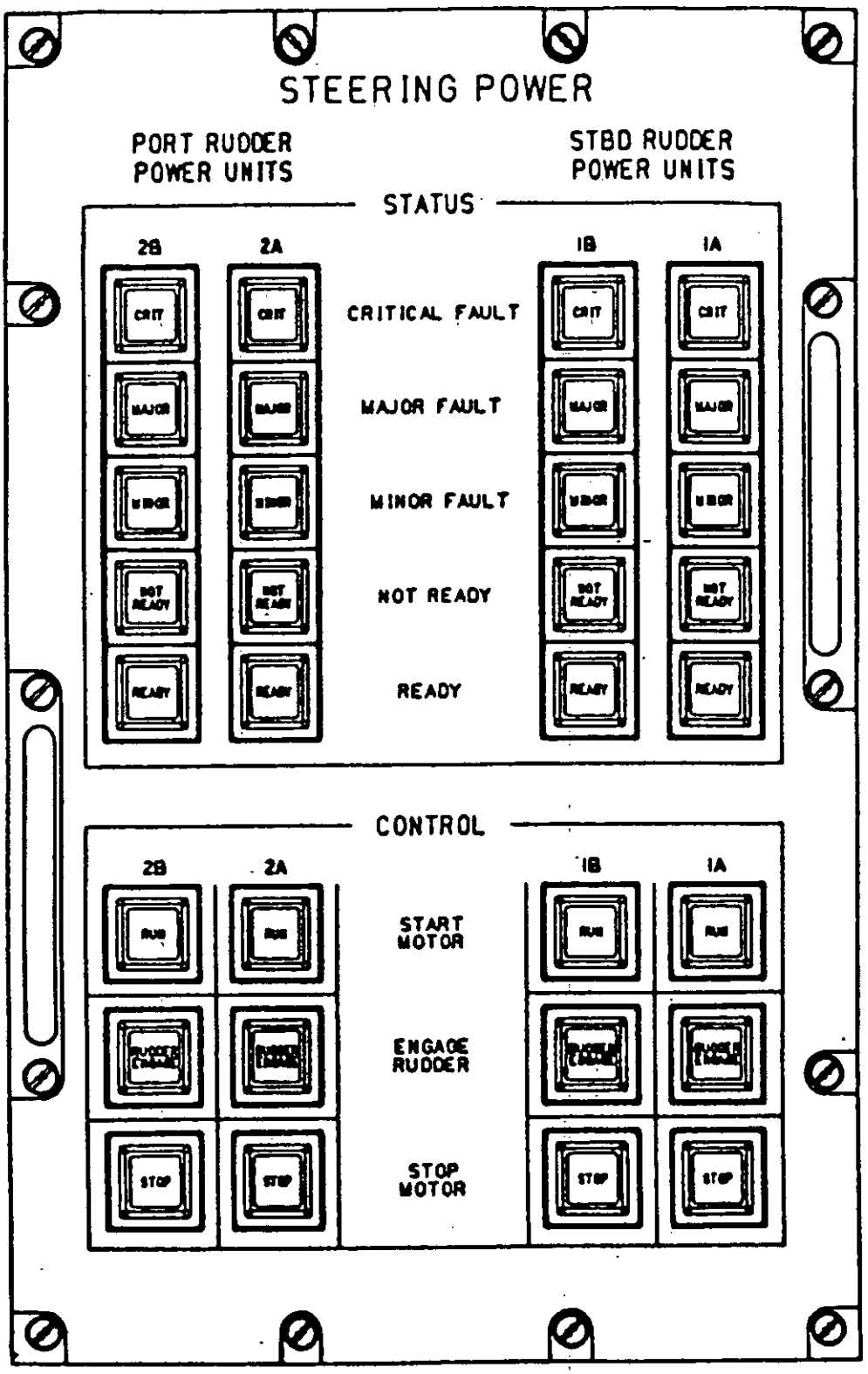


FIGURE 1. *Typical Steering Power Control Panel.*

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a. *CRITICAL FAULT.* There shall be a CRITICAL FAULT indicator for each hydraulic power unit. Each indicator shall illuminate when one or more of the following conditions are sensed in the associated hydraulic power unit:

(1) Low low oil level in head tank
(2) High high oil temperature in main pump case drain piping
(3) Electric motor overload
(4) Loss of ac power.

When one of the above conditions is sensed, the power unit shall automatically shut down, and the alternate power unit shall automatically be started and, if not already operating, put on-line. Each indicator shall be labeled to identify its corresponding hydraulic power unit.

b. *MAJOR FAULT.* There shall be a MAJOR FAULT indicator for each hydraulic power unit. Each indicator shall illuminate when one or more of the following conditions are sensed in the associated hydraulic power unit:

(1) Low oil level in head tank
(2) High oil temperature in supply tank
(3) Low servo pressure
(4) Servo control error
(5) Low replenishment pressure
(6) High motor temperature.

The low replenishment pressure sensor and the low servo pressure sensor shall be disabled during startup and after normal shutdown. Each indicator shall be labeled to identify its corresponding hydraulic power unit.

c. *MINOR FAULT.* There shall be a MINOR FAULT indicator for each hydraulic power unit. Each indicator shall illuminate when one or more of the following conditions exist:

(1) Return oil filter is dirty
(2) Servo oil filter is dirty
(3) Loss of one of the sets of electronics in the associated local control unit.

Each indicator shall be labeled to identify its corresponding hydraulic power unit.

d. *READY.* There shall be a READY indicator for each hydraulic power unit. When the Bridge Steering Unit is the controlling station, this indicator shall illuminate when all the following conditions exist:

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(1) AC power is applied to the associated local control unit.

(2) MODE SELECTOR switch in the associated motor controller is in the NORMAL position.

(3) No critical faults exist in the hydraulic power unit.

(4) MOTOR CONTROL SOURCE switch in the associated local control unit is in either the NORMAL or BRIDGE position.

(5) HELM ORDER SOURCE switch in the associated local control unit is in either the NORMAL or BRIDGE position.

(6) MODE selector switch in the associated local control unit is in the NORMAL position.

When the Aft Steering Unit is the controlling station, this indicator shall illuminate when all the following conditions exist:

(1) AC power is applied to the associated local control unit.

(2) MODE SELECTOR switch in the associated motor controller is in the NORMAL position.

(3) No critical faults exist in the hydraulic power unit.

(4) MOTOR CONTROL SOURCE switch in the associated local control unit is in either the NORMAL or AFT position.

(5) HELM ORDER SOURCE switch in the associated local control unit is in either the NORMAL or AFT position.

(6) MODE selector switch in the associated local control unit is in the NORMAL position.

e. *NOT READY.* There shall be a NOT READY indicator for each hydraulic power unit. When the Bridge Steering Unit is the controlling station, this indicator shall illuminate when one or more of the following conditions exist:

(1) No ac power to the associated local control unit.

(2) MODE SELECTOR switch in the associated motor controller is not in the NORMAL position.

(3) A critical fault exists in the associated hydraulic power unit.

(4) MOTOR CONTROL SOURCE switch in the associated local control unit is not in either the NORMAL or BRIDGE position.

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(5) HELM ORDER SOURCE switch in the associated local control unit is not in either the NORMAL or BRIDGE position.

(6) MODE selector switch in the associated local control unit is not in the NORMAL position.

When the Aft Steering Unit is the controlling station, this indicator shall illuminate when one or more of the following conditions exist:

(1) No ac power to the associated local control unit.

(2) MODE SELECTOR switch in the associated motor controller is not in the NORMAL position.

(3) A critical fault exists in the associated hydraulic power unit.

(4) MOTOR CONTROL SOURCE switch in the associated local control unit is not in either the NORMAL or AFT position.

(5) HELM ORDER SOURCE switch in the associated local control unit is not in either the NORMAL or AFT position.

(6) MODE selector switch in the associated local control unit is not in the NORMAL position.

f. RUN. There shall be a RUN indicator for each hydraulic power unit. It shall illuminate only when the associated electric motor main contactor is active. The indicators shall be labeled 1A, 1B, 2A, and 2B, as applicable.

g. STOP. There shall be a STOP indicator for each hydraulic power unit. It shall illuminate only when the associated electric motor is not running. The indicators shall be labeled 1A, 1B, 2A, and 2B, as applicable.

h. RUDDER ENGAGE. There shall be a RUDDER ENGAGE indicator for each hydraulic power unit. It shall illuminate only when the associated power unit is connected by the transfer valve to the rudder actuator. The indicators shall be labeled 1A, 1B, 2A, and 2B, as applicable.

3.2.3.3.2 Steering Mode Control Panel.

3.2.3.3.2.1 Physical characteristics. The panel height, including mounting flange, shall be not greater than 170 mm (6.5 inches). The panel width, including mounting flange shall be not greater than 320 mm (12.5 inches). The depth behind the panel, including connectors, shall be not greater than 200 mm (8 inches).

3.2.3.3.2.2 Functional characteristics. The following functional characteristics shall be provided in the Steering Mode Control Panel:

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a. Shall provide for control mode selection.

b. Shall provide for control transfer.

c. Shall provide a visual indication of which steering station has control of the steering.

d. Shall provide a visual indication of control mode.

e. Shall provide a visual indication of when the audible alarm has been muted (see 3.2.3.2.5).

f. Shall provide the means for acknowledging individual audible alarms (see 3.2.3.2.5).

g. Shall provide the means for testing the audible and visual indicators.

h. This equipment shall be interchangeable with all other Steering Mode Control Panels regardless of where they are mounted. Once mounted, the equipment shall operate properly for the steering station in which it is mounted without modification.

i. Shall provide System Status Information.

j. When compatibility with DMS is required (see 6.2), data communications with the SMCP in the BSU and each local control unit and the digital controller shall utilize DMS interfaces. When compatibility with DMS is not required data communications shall be as specified in 3.2.3.2.8.3.

k. Shall provide switches and indicators to support a heading monitoring feature. This feature, when selected, shall audibly and visually warn the helmsman when the current ship heading differs from the value of the ship heading when the feature was activated. The allowed difference (without triggering the alarm), shall be calculated as a function of ship mean heading and five times the average deviation from the mean, based on at least the last five minutes of heading data. As the average deviation from the mean during the reference period decreases, the limits which activate the alarm shall decrease. Independent of the algorithm, the alarm shall activate whenever the absolute deviation is greater than 10 degrees.

l. Shall provide switches and indicators to support a roll stabilization feature. This feature shall only be selectable in combination with either the Manual mode or the Automatic mode. When selected, the digital controller shall, in addition to meeting other requirements, minimize rudder movements that would induce or contribute to ship roll. This shall be accomplished by minimizing rudder movements whose frequencies are near the ship natural rolling frequency. If active rudder roll stabilization is specified (see 6.2), then the controller shall actively superimpose (add) rudder roll reduction movements to the rudder movement required for manual or automatic heading keeping.

3.2.3.3.2.3 Controls and switches. The following controls and switches shall be accessible at the front panel of the Steering Mode Control Panel (see figure 2 for typical panel layout).

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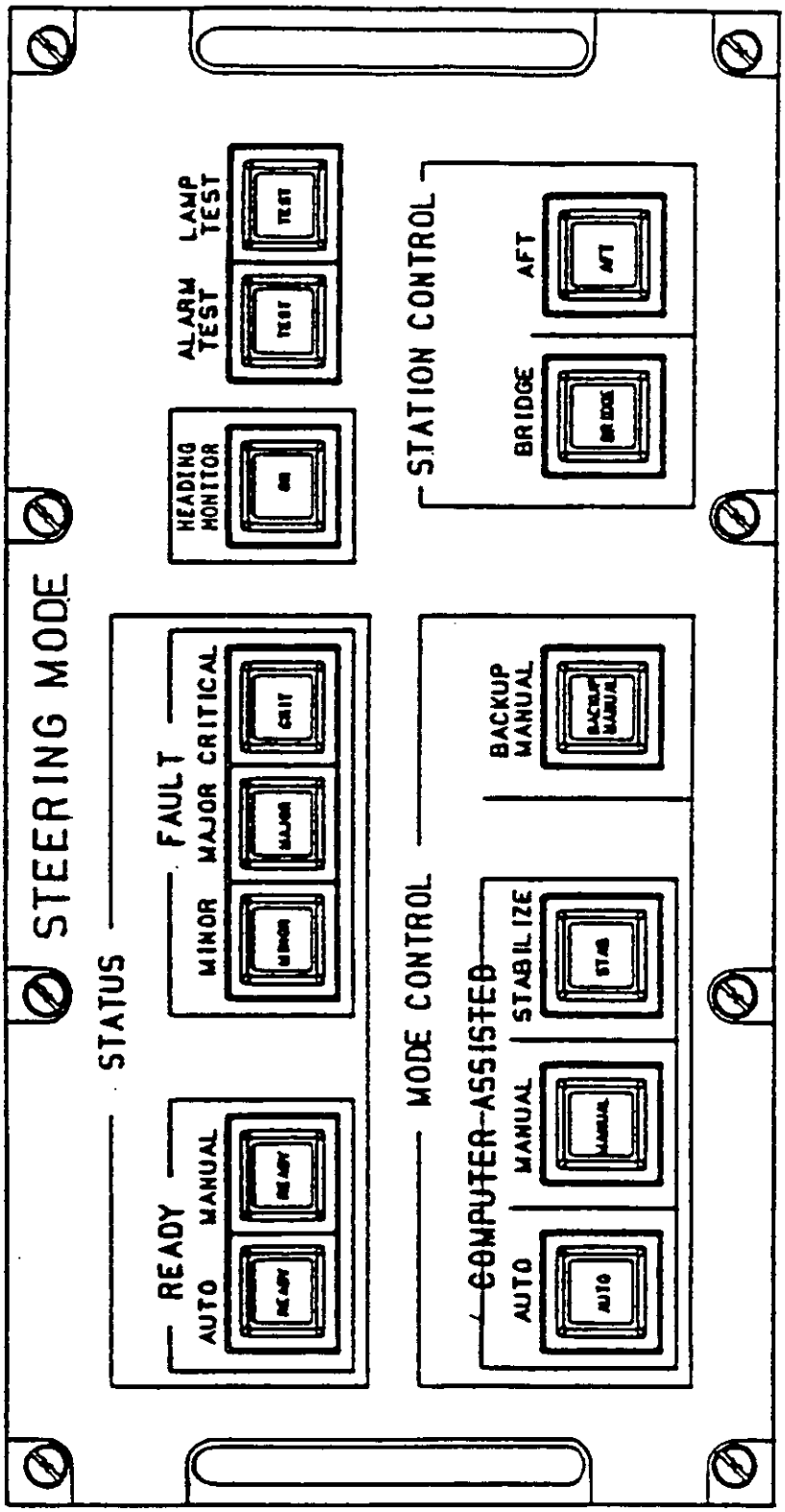


FIGURE 2. *Typical Steering Mode Control Panel.*

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a. *MODE CONTROL (AUTO).* This switch shall be used to select the Automatic (autopilot) mode of operation.

b. *MODE CONTROL (MANUAL).* This switch shall be used to select the Manual (steering wheel) mode of operation.

c. *MODE CONTROL (STABILIZE).* This switch shall be used to select operation with roll stabilization.

d. *MODE CONTROL (BACKUP MANUAL).* This switch shall be used to select the Backup Manual mode of operation.

e. *ALARM TEST.* This switch shall be used to test the operation of the audible alarm.

f. *LAMP TEST.* This switch shall be used to test all visual indicators.

g. *STATION CONTROL (AFT).* This switch shall be used to make a coordinated transfer of steering control from the Bridge Steering Unit to the Aft Steering Unit.

h. *STATION CONTROL (BRIDGE).* This switch shall be used to make a coordinated transfer of steering control from the Aft Steering Unit to the Bridge Steering Unit.

i. *ALARM MUTE.* These switches shall be combined with each fault indicator and, when pressed, shall mute the alarm associated with the particular fault indicator.

j. *HEADING MONITOR.* This switch shall be used to engage the heading monitor feature. It shall be a push-on, push-off type of switch. Pressing the switch during a heading monitor alarm condition shall both mute the audible alarm and turn off the heading monitor feature.

3.2.3.3.2.4 Indicators. The following indicators shall be mounted on the front panel of the Steering Mode Control Panel. When located in the Bridge Steering Unit, the brightness of each indicator shall be controlled by the master dimmer control.

a. *MANUAL (MODE CONTROL).* This indicator shall illuminate only when the Manual mode of operation has been selected. When operating in the Automatic mode, and the heading-keeping equipment malfunctions with the digital controller still operating properly or when steering control has switched to this mode from Automatic because of a reduction of ship speed to 2 m/s (4 knots) or below, this indicator shall flash, and an audible alarm shall sound to signal that the control mode has automatically changed from the Automatic mode to the Manual mode. To mute the alarm, the MODE CONTROL (MANUAL) button must be pressed. When pressed, this indicator shall display constant illumination.

b. *STAB (MODE CONTROL).* This indicator shall illuminate only when roll stabilization has been selected.

c. *AUTO (MODE CONTROL).* This indicator shall illuminate only when the Automatic mode of operation has been selected.

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d. *BACKUP MANUAL (MODE CONTROL).* This indicator shall illuminate when the Backup Manual mode of operation has been selected. When operating in either the Automatic or Manual mode and the digital controller malfunctions or when steering control has switched to this mode from Automatic because of a reduction of ship speed to 2 m/s (4 knots) or below, this indicator shall flash, and an audible alarm shall sound to signal that the control mode has automatically changed to the Backup Manual mode. To mute the alarm, the BACKUP MANUAL (MODE CONTROL) button must be pressed. When pressed, this indicator shall display constant illumination.

e. *STATION CONTROL (AFT).* This indicator shall illuminate only when the Aft Steering Unit is controlling the steering. It shall alternately turn on and off when a request for steering control is initiated at the Aft Steering Unit, or when the Bridge Steering Unit requests that control of the steering be transferred from the Bridge Steering Unit to the Aft Steering Unit.

f. *STATION CONTROL (BRIDGE).* This indicator shall illuminate only when the Bridge Steering Unit is controlling the steering. It shall alternately turn on and off when a request for steering control is initiated at the Bridge Steering Unit or when the Aft Steering Unit requests that control of the steering be transferred from the Aft Steering Unit to the Bridge Steering Unit.

g. *READY (STATUS – AUTO).* This indicator shall illuminate when the autopilot equipment is fully operational and available for use.

h. *READY (STATUS – MANUAL).* This indicator shall illuminate when the Manual mode of operation is fully operational and available for use.

i. *MINOR (STATUS – FAULT).* This indicator shall illuminate when the digital controller detects a fault condition which does not alter full performance of any mode, but does require that the operator be alerted to the faulting condition. This indicator shall illuminate when the steering wheel at any station is not in use and is not set to 0 ± 5 degrees. This warns of a potentially hazardous condition which will result if a transfer to manual control at any station is made when the steering wheels are not set to zero degrees. The audible alarm shall not be activated with the detection of minor faults.

j. *MAJOR (STATUS – FAULT).* This indicator shall illuminate when the digital controller detects a fault which prevents full performance operation in any control mode.

k. *CRIT (STATUS – FAULT).* This indicator shall illuminate when a fault is detected that will contribute to the generation of an erroneous rudder angle order signal.

l. *ON (HEADING MONITOR).* This indicator shall illuminate only when the heading monitor function has been selected. It shall flash under the alarm conditions specified in 3.2.3.3.2.2.k.

3.2.3.3.3 Automatic Steering Control Panel.

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3.2.3.3.3.1 Physical characteristics. The panel height, including mounting flange, shall be not greater than 360 mm (14 inches). The panel width, including mounting flange, shall be not greater than 230 mm (9 inches). The depth behind the panel, including connectors shall be not greater than 200 mm (8 inches).

3.2.3.3.3.2 Functional characteristics. The following functional characteristics shall be provided in the Automatic Steering Control Panel:

a. Shall provide the means for entering ordered heading data into the system.

b. Shall provide the means for entering the performance setting into the system.

c. Shall provide the means for entering rudder limit settings into the system.

d. Shall provide a digital indication of the following information:

(1) Ship heading
(2) Ordered heading
(3) Keypad value
(4) Rudder limit setting.

e. Shall provide the means to quickly and simply order full left or right rudder without switching modes. When this feature is selected, a new ordered heading shall be continuously calculated and displayed by the digital controller. This calculation shall be a prediction of what the steady-state heading would be if the rudder was required to reduce the ship turning rate to zero in minimum time. The digital controller shall execute this maneuver when the helmsman releases the LEFT TURN or RIGHT TURN push button.

f. All indicators and displays shall be extinguished when the heading-keeping equipment is inoperable for any reason.

g. This equipment shall be interchangeable with all other Automatic Steering Control Panels regardless of where they are mounted. Once mounted, the equipment shall operate properly for the steering station in which it is mounted without modification.

h. When compatibility with DMS is required (see 6.2), all data communications between the ASCP in the BSU and the local control units shall utilize DMS STANAG 4156 (protocol type A) interfaces. When compatibility with DMS is not required, data communications shall be as specified in 3.2.3.2.8.3.

i. Shall provide the means to specify the direction the ship is to turn to obtain a new ordered heading.

3.2.3.3.3.3 Controls and switches. The following controls and switches shall be accessible on the front panel of the Automatic Steering Control Panel (see figure 3 for typical panel layout).

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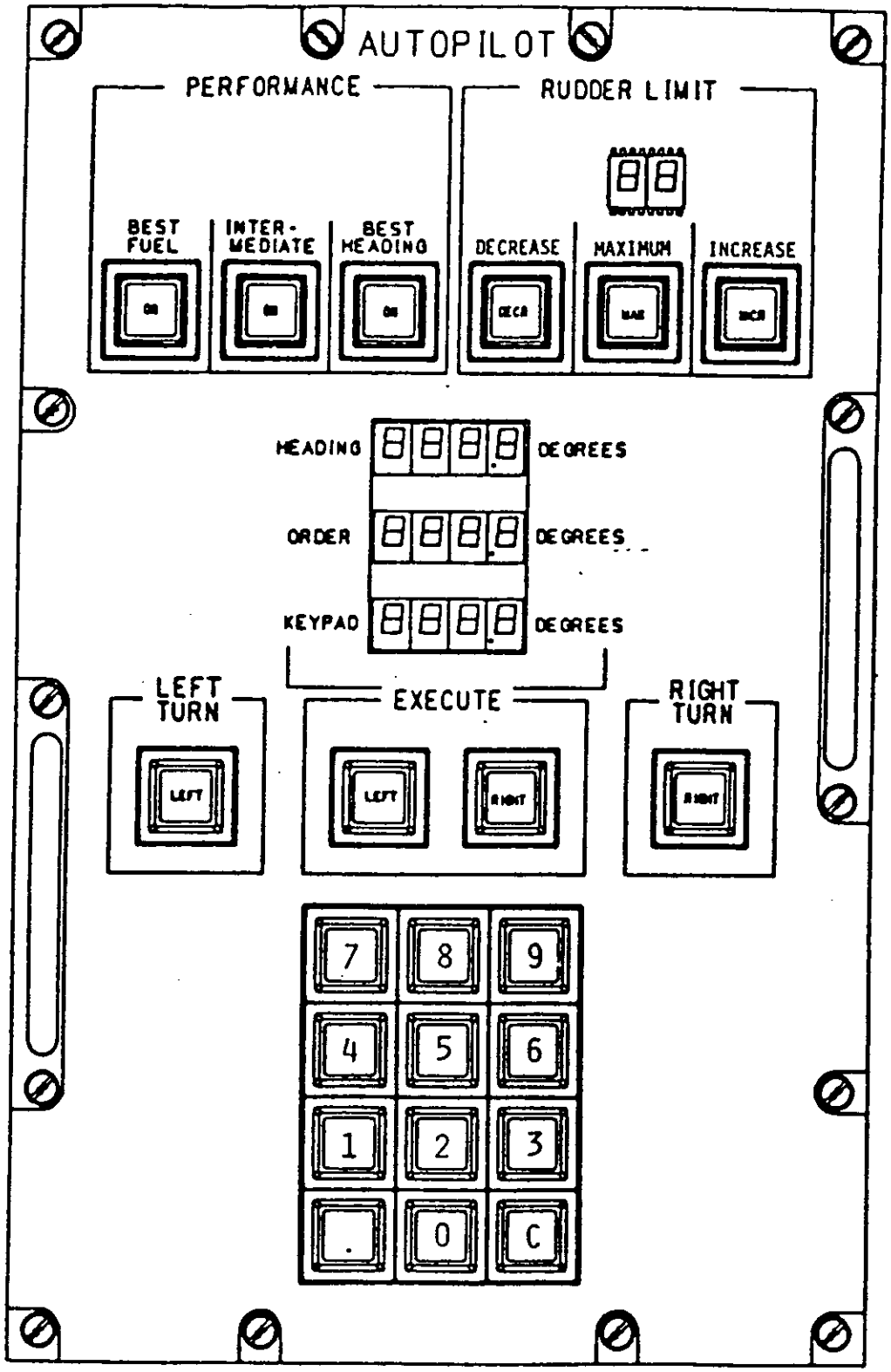


FIGURE 3. *Typical Automatic Steering Control Panel.*

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a. *PERFORMANCE (BEST FUEL).* This switch shall be used to select the lowest controller gain as the lower limit of the adaptive search.

b. *PERFORMANCE (BEST HEADING).* This switch shall be used to select the highest controller gain. In this mode there shall be no adaptive search.

c. *PERFORMANCE (INTERMEDIATE).* This switch shall be used to select a gain value intermediate between lowest and highest as the lower limit of the adaptive search.

d. *HEADING ORDER INPUT KEYPAD.* This control shall be used to enter heading commands to the autopilot circuitry. A keypad having keys labeled 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., and C shall be used to enter heading data into the autopilot. Successive depression of these keys shall enter data into the keypad register, and the data shall be entered into the digital controller when the EXECUTE (LEFT) or EXECUTE (RIGHT) button is pressed.

e. *EXECUTE (LEFT).* This push button shall be used to enter the numerical data in the keypad register into the digital controller as ordered heading, and cause the ship to turn left to attain that heading.

f. *EXECUTE (RIGHT).* This push button shall be used to enter the numerical data in the keypad register into the digital controller as ordered heading, and cause the ship to turn right to attain that heading.

g. *RUDDER LIMIT (INCREASE).* This switch shall be used to increment the rudder limit command. The rudder limit command shall set the maximum rudder travel from hard-over to hard-over in degrees. Depression of this switch shall increment the rudder limit command at a rate of 1.0 degree per second until the switch is released. The value indicated on the rudder limit display when the switch is released shall be used as the rudder limit command. This switch shall have a legend reading INCREASE.

h. *RUDDER LIMIT (DECREASE).* This switch shall be used to decrement the rudder limit command. Depression of this switch shall decrement the rudder limit command at a rate of 1.0 degree per second until the switch is released. The value indicated on the rudder limit display when the switch is released shall be used as the limit command. This switch shall have a legend reading DECREASE.

i. *RUDDER LIMIT (MAXIMUM).* This switch shall be used to set the rudder limits to the maximum value. This switch shall have a legend reading MAXIMUM.

j. *LEFT TURN.* This switch shall be used to order an immediate excursion of the rudders to the selected rudder limit for a turn to the left. When the switch is released, the system shall maintain the calculated ordered heading.

k. *RIGHT TURN.* This switch shall be used to order an immediate excursion of the rudders to the selected rudder limit for a turn to the right. When the switch is released, the system shall maintain the calculated ordered heading.

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3.2.3.3.3.4 Indicators and displays. The following indicators shall be mounted on the front panel of the Automatic Steering Control Panel. When located in the Bridge Steering Unit, the brightness of each indicator shall be controlled by the master dimmer control.

a. *RUDDER LIMIT.* This display shall display the selected limit of rudder travel from hard-over to hard-over. The display shall be digital displaying a range of 5 to 35 degrees in 1.0-degree increments. The height of each digit shall be not less than 6.4 mm (0.25 inch). This display shall be active regardless of which steering unit is in control.

b. *HEADING.* This display shall display the prevailing heading of the ship in degrees. The display shall be digital and display a range of 0 to 360 degrees in 0.1-degree increments. The height of each digit shall be not less than 6.4 mm (0.25 inch). The display shall be active regardless of which steering unit is in control.

c. *ORDER.* This display shall display the ordered heading to be used by the autopilot circuitry. It shall be digital and display a range of 0 to 360 degrees in 0.1-degree increments. The height of each digit shall be not less than 6.4 mm (0.25 inch). This display shall be active regardless of which steering unit is in control.

d. *KEYPAD.* This display shall display the contents of the keypad register. It shall be digital and display a range of 0 to 360 degrees in 0.1-degree increments. The height of each digit shall be not less than 6.4 mm (0.25 inch). This display shall be active only when steering is being controlled from the associated Automatic Steering Control Panel.

e. *LEFT (TURN).* This indicator shall illuminate when the ordered heading has been changed in such a manner as to cause the ship to execute a left turn. The ordered heading can be changed with either the LEFT TURN switch or the keypad. The indicator shall be extinguished when the ship heading equals the ordered heading or a new ordered heading is entered which would cause the ship to enter a right turn.

f. *RIGHT (TURN).* This indicator shall illuminate when the ordered heading has been changed in such a manner as to cause the ship to execute a right turn. The ordered heading can be changed with either the RIGHT TURN switch or the keypad. The indicator shall be extinguished when the ship heading equals the ordered heading or a new ordered heading is entered which would cause the ship to enter a left turn.

3.2.3.3.4 Helm angle indicator. This unit shall indicate the actual position of the steering wheel. It shall be an analog indicator mechanically linked to the shaft of the steering wheel assembly.

3.2.3.3.5 Emergency Override to Manual Panel. This unit shall incorporate a push button switch with a diameter of not less than 50 mm (2 inches), which when pressed shall effect an immediate transfer of control to the station where the button is pressed. The control subsystem shall immediately revert to the Backup Manual mode of operation. If this button is pressed while the digital controller and digital electronics are operational, the digital controller shall relinquish control of the steering but shall continue its monitoring function. When compatibility with DMS is required (see 6.2), all data communications between the Emergency Override to Manual Panel in the ship control console and other units of the control system shall utilize the DMS discrete interfaces.

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3.2.3.3.6 Manual steering wheel assembly. The manual steering wheel assembly shall have a steering wheel with a diameter of 300 mm (12 inches) and make 12 revolutions for hard-over to hard-over rudder movement. The steering wheel shall be clutched to permit overtravel at the end points without damaging the mechanism. The assembly shall be mechanically connected to four precision potentiometers and one synchro. Each potentiometer and synchro shaft shall rotate a total of 300 degrees for hard-over to hard-over rotation of the wheel. Each shaft shall track the rotational position of the other shafts within 0.2 degree throughout the range of rotation. The potentiometers shall be in accordance with MIL-R-39023 and the synchro shall be in accordance with MIL-S-20708. Each potentiometer shall have a direct current (dc) power source to supply its excitation voltage, and this power supply shall be located in the associated local control unit. When compatibility with DMS is required (see 6.2), the synchro signals from the manual steering wheel assembly in the ship control console shall be transmitted via DMS.

3.2.3.3.7 Local control unit (LCU).

3.2.3.3.7.1 Physical characteristics. Each local control unit shall protrude less than 300 mm (12 inches) from the bulkhead. The width shall be less than 560 mm (22 inches) and the height shall be less than 560 mm (22 inches).

3.2.3.3.7.2 Functional characteristics. Each local control unit shall have the following functional characteristics:

a. The unit shall contain the necessary electronic circuitry for controlling the servo valve in its associated hydraulic power unit. The circuitry shall be duplicated so that one set of electronics controls one of the two coils in the servo valve and the duplicate set is dedicated to the other coil. Each set of electronic circuitry shall have a separate power supply. The active coil shall be energized by one set of electronics while the other set of electronics drives a dummy load. A switch shall be provided to manually select which set of electronics will energize the coil. This selection shall override digital controller control.

b. Shall provide the means for manually selecting the source of rudder angle order signals.

c. Shall be powered from the associated motor controller. At no time shall the local control unit have 115-volts ac (Vac) power, unless the associated motor controller has 115-Vac control power.

d. Shall provide the means to manually select the signal source for controlling the electric motor in the associated hydraulic power unit.

e. Shall provide the means to manually take the local control unit off-line and place it into a maintenance mode that utilizes the digital controller to assess local control unit operability.

f. Each local control unit shall be interchangeable with all other local control units regardless of where they are mounted. Once mounted, the equipment, without modification, shall operate properly for the position in which it is located.

g. Shall provide the means to manually disable all control by the digital controller.

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h. Circuitry to cause an automatic transfer between the hydraulic power units that operate one rudder can be located either in the motor controller or the local control unit. For characteristics of this automatic transfer, refer to the main pump motor and controller requirements of MIL-S-17803.

i. Shall contain circuitry to provide digital to analog conversion, amplification, and filtering of signals.

j. Shall contain circuitry to provide analog to digital conversion, scaling, and filtering of signals.

3.2.3.3.7.3 Controls and switches. The following controls and switches shall be accessible at the front panel (behind door) of each local control unit, and shall be labeled as indicated:

a. *HELM ORDER.* This control shall be used to internally generate ordered rudder angle signals. The control shall be a single-turn, precision potentiometer.

b. *HELM ORDER SOURCE.* This switch shall be used to select the source or rudder angle order commands. It shall have positions labeled LCU 2B, LCU 2A, LCU 1B, LCU 1A, AFT, BRIDGE, and NORMAL, as applicable. When in the NORMAL position, the rudder angle order commands selected shall depend upon the control mode selected and the location of the controlling station. When in the BRIDGE position, the rudder angle order command shall be selected directly from the Bridge Steering Unit helm order transmitter. When in the AFT position, the rudder angle order command shall be selected directly from the Aft Steering Unit helm order transmitter. When in the one of the LCU positions, the rudder angle order command shall be selected from the helm order control associated with the selected LCU.

c. *AMPLIFIER MODE.* This three-position switch shall be used to select the mode of operation of the dual set of servo electronics. When in the NORMAL position, the active servo electronics shall be determined by digital controller evaluation. When in the CH A position, the A set of servo electronics shall be active while the B set of servo electronics drives a dummy load. When in the CH B position, the active servo electronics shall be the B set, and the A set shall drive a dummy load.

d. *DIGITAL CONTROL.* This two-position switch shall be used to disable all control of the local control unit by the digital controller. When in the ON position, control by the digital controller shall be enabled. When in the OFF position, control by the digital controller shall be disabled.

e. *MOTOR CONTROL SOURCE.* This switch shall be used to select the signal source for controlling the electric motor in the hydraulic power unit. It shall have three positions, NORMAL, BRIDGE, and AFT. When in the NORMAL position, motor control and rudder engagement from either of the two remote controlling stations shall be permitted. When in the BRIDGE position, remote motor control and rudder engagement shall be permitted only at the Bridge Steering Unit. When in the AFT position, remote motor control and rudder engagement shall be permitted only at the Aft Steering Unit.

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f. *MODE.* This switch shall be used to take the local control unit off-line and place it into a maintenance mode. The switch shall have two positions, NORMAL and TEST. When in the NORMAL position, the local control unit shall be ready for operation on-line. When in the TEST position, the local control unit shall be taken off-line and placed in a maintenance mode.

3.2.3.3.7.4 Test mode. Each local control unit shall have a test mode which, when selected, shall enable the digital controller to evaluate the operability of the unit. This off-line mode shall provide the following characteristics:

a. *Isolation.* A manually operated switch shall be provided to select the test mode, and this switch shall isolate all control lines entering or exiting the local control unit which could affect system operation.

b. *Stimuli interfacing.* The local control unit shall accept signals from the digital controller which will simulate start and stop motor commands, engage rudder commands, emergency override commands, sensor signals from the companion hydraulic power unit, rudder angle order commands, and critical fault signals.

c. *Response monitoring.* The local control unit responses to digital controller-generated stimuli shall be monitored by the digital controller.

3.2.3.3.8 Rudder angle transmitter unit.

3.2.3.3.8.1 Physical characteristics. The housing of each rudder angle transmitter unit (RATU) shall have a length less than 250 mm (10 inches), a width less than 250 mm (10 inches), and a height less than 250 mm (10 inches).

3.2.3.3.8.2 Functional characteristics. Each rudder shall have two rudder angle transmitter units which will supply the control subsystem with electrical signals proportional to the actual position of the rudder. Each transmitter unit shall have the following characteristics:

a. Shall contain dual potentiometers in accordance with MIL-R-39023, with provision for mechanically nulling the potentiometers. Such provision shall be readily accessible when the RATU is mounted in place on the ship.

b. Shall contain one each type 15CX6C synchro in accordance with MIL-S-20708, with provision for mechanically nulling the synchro. Such provision shall be readily accessible when the RATU is mounted in place on the ship.

c. The units shall be located approximately opposite each other on the rudder stock.

d. Each unit shall be driven by a dedicated linkage that includes a means to adjust the linkage to form a perfect parallelogram.

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e. Excitation voltages for the potentiometers shall be provided by the associated local control unit.

f. Synchro signals shall not be placed in the same connector with potentiometer signals.

3.2.4 Physical. In the ship spaces where steering equipment shares space with other ship systems, such as the bridge, the steering equipment shall be enclosed in functionally grouped modular units to facilitate integration into larger consoles or equipment racks. The Aft Steering Unit shall weigh less than 1,800 N (400 pounds) and protrude less than 760 mm (30 inches) from the bulkhead. The width shall be less than 1,220 mm (48 inches) and the height shall be less than 1,520 mm (60 inches).

3.2.5 Reliability. Reliability terms, as used herein, are defined in MIL-STD-721. Each unit of the steering control system shall meet the following quantitative reliability requirements (see 6.3):

a. Each unit shall have a mean time between failures (MTBF) of not less than 5,000 hours, except that the Aft Steering Unit shall have a MTBF of not less than 2,500 hours.

b. Elements of the system considered to be nonrepairable shall have a MTBF of not less than 20,000 hours.

3.2.6 Maintainability. Each steering unit of the steering control system shall meet the following quantitative maintainability requirements (see 6.3):

a. The mean time to repair (MTTR) for each steering unit shall be not greater than 1/2 hour for all repair actions performed at the organizational level of maintenance.

b. Not less than 90 percent of the repair actions performed at the organizational level of maintenance shall be accomplished within 1 hour.

3.2.7 Environmental conditions. Each unit of the control system shall operate continuously and reliably under the environmental conditions specified herein. Performance shall not be degraded due to operation under any environmental conditions specified herein. Following exposure to environmental conditions specified herein, there shall be no deterioration, corrosion, or electrical measurement not within the specified tolerance limits.

3.2.7.1 Temperature. Each major component of the control system shall withstand the operating and nonoperating temperature extremes as specified herein.

3.2.7.1.1 Low temperature. Each unit of the control system shall operate continuously without damage or performance degradation in an ambient temperature of 0 degrees Celsius (°C) (+0, −5 °C).

3.2.7.1.2 High temperature. Each unit of the control system shall operate continuously without damage or performance degradation in an ambient temperature of +50 (+5, −0 °C).

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3.2.7.1.3 Storage temperature. Each unit of the control system shall not be damaged or suffer performance degradation when restored to the operating temperature range after being exposed to the nonoperating temperature range from −55 (+0, −5) to 80 (+5, −0) °C.

3.2.7.2 Humidity. Each unit of the control system shall withstand the effects of humidity up to 95 percent, including conditions wherein condensate develops in and on the unit.

3.2.7.3 Vibration. Each unit of the steering control system shall withstand the type I vibration test in accordance with MIL-STD-167-1, from 4 hertz (Hz) (or lowest attainable frequency) up to and including 50 Hz. After completion of the vibration test, each unit shall withstand a dielectric strength test voltage of 325 V without damage.

3.2.7.4 Shock. Each unit of the steering control system shall withstand a grade A, class I, type A, high-impact shock in accordance with MIL-S-901 (see 4.6.20).

3.2.7.5 Noise. The steering control system shall meet the requirements specified herein when tested in accordance with 4.6.15. Steering control system equipment located in the bridge shall not exceed the grade A12 airborne sound level acceptance criteria in accordance with MIL-STD-740-1. Equipment in the steering gear room shall not exceed the grade D airborne sound level acceptance criteria in accordance with MIL-STD-740-1. Steering control system equipment shall not exceed the type III structureborne noise acceptance criteria in accordance with MIL-STD-740-2.

3.2.7.6 Pitch and roll. The control system shall not be damaged or malfunction when exposed to pitch and roll at any angle up to 45 degrees from the vertical. No favorable orientation may be used.

3.2.7.7 Inclination. The control system shall not be damaged or malfunction when permanently inclined at any angle up to 15 degrees from the vertical in any direction. No favorable orientation may be used.

3.2.8 Transportability. Each unit of the control system shall be transportable by rail, barge, truck, or ship without deterioration, degradation, or damage. General environmental considerations for transportation of the equipment shall be in accordance with SAE HIR 1063. Each unit of the control system shall not be damaged by air transportation at altitudes up to 15.25 km (kilometers) (50,000 feet) in nonpressurized compartments.

3.2.9 Digital controller requirements.

3.2.9.1 Digital controller. The digital controller shall perform the instruction repertoire necessary to control the memory, input/output, and the applicable digital interfaces to effect control of the steering system. The digital controller shall be augmented by the built-in test equipment (BITE) electronics and the control/maintenance panel.

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3.2.9.2 Memory. The memory shall contain read-only memory (ROM) and random access memory (RAM) memory arrays. The ROM memory shall contain the necessary algorithms to perform the steering function. The RAM memory shall serve as a scratch pad memory and as a temporary storage device.

The memory interface shall transfer memory addresses and data between the memory and the digital controller. The memory shall contain the control logic and signals required to handshake with the memory interface.

Upon power-on reset, the digital controller shall execute a routine, stored in the read-only memory, which shall ensure a safe startup regardless of the shutdown method.

3.2.9.3 Nonreturn-to-zero (NRE) digital interfaces. When compatibility with DMS is required (see 6.2), the digital controller shall have three STANAG 4156 Digital Interfaces: DMS, the LCUs, and the Aft Interface Electronics Unit.

3.2.9.3.1 Data multiplex system interface. The interface to the DMS shall be performed via Input/Output Channel Adapters (IOCA) and shall conform to ANEP-4, NATO STANAG 4156 (protocol B). The interface shall be accomplished with two redundant channels.

3.2.9.3.2 Local control unit interface. The interface to the LCU shall be performed via IOCAs and shall conform to NATO STANAG 4156 (protocol A). The interface shall be accomplished with a single channel to each LCU.

3.2.9.3.3 After interface electronics unit (AIEU). The interface to the AIEU shall be performed via IOCAs and shall conform to NATO STANAG 4156 (protocol A). The interface shall be accomplished with a single channel to the AIEU.

3.2.9.4 Built-in-testing (BIT) requirements. The BIT shall provide automatic test capability to detect, localize and isolate faults. The primary objective shall be to fault detect and isolate to the line replaceable unit (LRU). BIT shall achieve detectability and fault isolation levels sufficient to meet specified MTTR requirements. BIT electronics shall be functional on line and off line. BIT electronics shall detect digital controller non-recoverable failure and shall provide the signals necessary to cause the associated reversion to backup manual mode.

3.2.9.4.1 Fault definition. A fault shall be defined as a degradation in performance due to detuning, maladjustment, misalignment, failure of part or parts, and the like.

3.2.9.4.2 BIT fault-isolation level. At least 95 percent of all system failure and out-of-tolerance modes, for all subassemblies, shall be isolated and identified with the replacement of the first five or fewer standard electronic modules (SEMs) or LRUs of a fault group by means of the BIT circuitry and indicators and without aid of any external test equipment.

3.2.9.4.3 BIT false-alarm rate. The mean time between false indication of equipment status, considering both the system equipment and BIT, shall be greater than 1 percent of the MTBF.

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3.2.9.4.4 Fail-safe provisions. The circuits and devices which provide BIT and fault-isolation functions shall be designed so that failure of these circuits and/or devices will not cause failure of the steering control subsystem. BIT circuitry shall not interfere with the operation of the prime equipment.

3.2.9.4.5 BIT procedures. BIT shall perform a self-test before testing control subsystem hardware. BIT procedures shall be automatically initiated. Automatically initiated tests shall be executed at least 12 times per hour, and shall not interfere with normal control subsystem operation. A means to manually initiate BIT procedures shall be provided. Operation of self-test equipment shall not require removal or disconnection of steering control subsystem components.

The BIT electronics shall provide the capability through the control and maintenance panel to identify the sensed condition which initiated the critical, major, or minor fault indication. The summary fault indicator on the control and maintenance panel shall allow the operator to identify the condition after it has cleared.

3.2.9.5 Control and maintenance panel. The control and maintenance panel shall contain the logic and control signals required to interface the panel to the digital controller.

3.2.9.5.1 Physical characteristics. The maintenance panel shall be sized so that it can be mounted in the Aft Steering Unit. Portions of the control and maintenance panel, except for the summary fault indicator, shall have limited accessibility. The summary fault indicator shall be exposed to compartment personnel at all times.

3.2.9.5.2 Functional characteristics. The following functional characteristics shall be provided by the maintenance panel:

a. Display BIT results.

b. Display descriptions of current steering system faults.

c. Control BIT and display operation as required.

d. Provide digital controller manual reset capability.

e. Provide a summary control system fault indicator.

f. The panel shall be furnished with an alphanumeric display of not less than 6 lines of 40 characters each. A readily accessible switch shall be provided to manually activate and deactivate the display. A feature shall be provided which will automatically deactivate the display following a period of 5 minutes of inactivity.

g. The panel shall display fault isolation data in an English format that does not require the use of reference or look-up tables.

3.2.9.6 Communication. Inter-unit communication hardware and software shall be structured into hierarchical levels. Levels shall be:

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a. **Level 1:** *Protocol.* Level 1 shall handle the hardware interface, perform the interface protocol function, and perform word-and message-level error checking. Level 1 shall include, as appropriate, generation and checking of parity bits, channel addresses, and check sums.

b. **Level 2:** *Channel control.* As appropriate, Level 2 shall perform multi-message-level error checking by using the following error checking techniques as a minimum:

(1) *Sequence numbering.* Each message exchanged between control system units shall be assigned a sequence number. Received messages with unexpected sequence numbers shall be considered to be in error.

(2) *Acknowledgement.* Successful receipt by a control subsystem unit of an information message (any message except an acknowledgement message) shall cause the receiving unit to send a positive acknowledgement message to the transmitting unit. If a control subsystem unit receives an invalid information message but can determine the message's transmitter, the unit shall send a negative acknowledgement to that transmitter. Messages that are negative acknowledged shall be considered to be in error. Acknowledgement messages shall include the sequence number of the message being acknowledged.

(3) *Timeout.* Transmitted messages, other than acknowledgement messages, shall be considered to be in error if they are not acknowledged within a reasonable time period. Control subsystem units shall be considered to be in error if no messages are received from them within a reasonable time period.

Level 2 shall use channel clear messages for initialization of channels as required for error recovery and power-on reset procedures. It shall use test messages to periodically test the working/non-working status of inter-unit communication paths. When two channels are available for inter-unit communication, each unit shall send and receive messages via both channels. Messages shall be assigned to channels as necessary to equalize traffic. If a channel fails, it shall not be used.

c. **Level 3:** *Information exchange.* As appropriate, Level 3 shall handle control system inter-unit information exchange. Level 3 shall use status change messages to inform other units of changes in the status of discrete parameters. It shall use information request messages to transmit requests for information. It shall use information response messages to respond to previously transmitted information request messages.

Message routing. Information request messages shall be sent to the control system unit which senses or computes the requested information. Information response messages shall be sent to the control system unit which originated the information request message that is being responded to.

3.2.10 Software requirements. Software requirements shall be as specified in 3.2.10.1 through 3.2.10.3 (see 6.3).

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3.2.10.1 Application programs. Application programs shall be written in ADA. Programming shall be in accordance with the standards, conventions, and procedures detailed in DOD-STD-2167. Application program identification, testing, and quality assurance shall be in accordance with DOD-STD-2167. SDEX/M, SDEX/20, or other approved executives shall be used as specified (see 6.2). Configuration management practices shall be in accordance with MIL-STD-483. Specification practices shall be in accordance with MIL-STD-490.

3.2.10.2 Modules. Application programs shall be divided into modules. Modules shall be designed to facilitate independent coding and testing, and for application to multiple control system units when applicable. Modules shall include as a minimum:

a. **Module EF:** *Executive Functions.* Module EF shall contain the code necessary to initiate execution of the program and to schedule tasks.

b. **Module IC:** *Inter-unit Communication.* Module IC shall contain the code necessary to receive and transmit inter-unit messages.

c. **Module OI:** *Operator Interface.* Module OI shall contain the code necessary to interface the operator's controls and displays with the global data base.

d. **Module PI:** *Plant Interface.* Module PI shall contain the code necessary to interface monitored and controlled machinery and systems with the global data base.

e. **Module SC:** *Steering Control.* Module SC shall contain the code necessary to implement the control features associated with the electrohydraulic steering equipment.

f. **Module SD:** *Status Detection.* Module SD shall contain the code necessary to detect changes in the status of discrete signals. The resulting list of changed signals shall be used by other modules to initiate appropriate actions.

g. **Module SM:** *System Monitoring.* Module SM shall contain the code necessary to execute control system built-in test procedures.

3.2.10.3 Reserve margin. Each digital controller shall have memory input/output capability, and execution time reserves of not less than 20 percent.

3.2.11 Interior communications system interfaces. The steering control system shall operate from gyrocompass and ship speed signals available on the ship.

3.3 Construction. Construction shall be as specified in 3.3.1 through 3.3.11 (see 6.3).

3.3.1 Electrical.

3.3.1.1 Primary power supply. The steering control system shall operate and deliver rated performance from the type I power supply characteristics in accordance with DOD-STD-1399, section 300, when operating under the allowable power supply variations specified therein. The nominal power supply voltage rating for the control system shall be 115 Vac (rms), 3-phase, in accordance with

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type I power of DOD-STD-1399, section 300. The control system shall be rated for nominal frequency of 60 hertz. The control system shall not affect the type I power supply by distorting any of the characteristics.

3.3.1.1.1 Transients. The control system shall meet the requirements of this specification during the period of the voltage and frequency transients specified in DOD-STD-1399, section 300, for type I power. The transients shall not cause failure of any part or prevent resumption of normal operation, or require the equipment to recycle when the transients have ceased, or require recalibration, resetting, or adjustment to ensure proper operation of the control system after these transients.

3.3.1.1.2 Overvoltage. The control system shall be capable of being continuously energized for a period of 12 hours from a 60-Hz source with the voltage equal to 149 Vac (rms), with the control system at an ambient temperature of +50 (+5, −0) °C without failure of any parts. During this period the control system is not required to meet the performance accuracies specified herein; however, after nominal voltage has been restored, these accuracy requirements shall be met.

3.3.1.2 Electrical overload protection. Unless otherwise specified herein, the control system shall conform to requirement 8 of MIL-STD-454 for class 1 equipment.

3.3.1.3 Wiring methods.

3.3.1.3.1 Harnessing. Wiring shall be neatly formed into groups that shall be laced, tied, or clamped in a manner that provides support and prevents chafing of the wire insulation resulting from vibration and shock. There shall be no splices in the wire, and connections shall be made at the terminals of the devices or at a terminal block.

3.3.1.3.2 Slack. Sufficient slack shall be provided so as not to impair movement or put undue stresses on the wire or components in those places where movement of components may be executed. Slack shall also be provided to prevent undue stress on terminal connections resulting from shock and vibration. Where wire terminals are used, sufficient slack shall be provided for at least two replacements of the part in the event that the wires are damaged or have to be clipped at the terminals during disassembly.

3.3.1.3.3 Mechanical supports. Electrical connections shall be provided with supports to prevent breakage and minimize changes in performance resulting from shock, vibration, and inclination.

3.3.1.3.4 Clamp contact connections. Electrical connections shall not be dependent on wires, lugs, terminals, or similar components clamped between a metallic member and an insulating pliable material. Such connections shall be clamped between metal members.

3.3.1.3.5 Insulation protection. Wherever wires are run through holes in metal partitions, the holes shall be equipped with grommets for mechanical protection of insulation. Care shall be exercised in the running of wires to ensure that they are not carried over or bent around any sharp corner or edge.

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3.3.1.3.6 Shielded wire. Conductors using metallic shielding unprotected by an outer insulation shall not be used. Shielding shall be terminated as close to the circuit terminal as practical without risk of grounding the terminal. Where required by circuit design, shielding shall be grounded and the connection to the chassis shall be made by means of terminal lugs on screw-type terminals or bolts.

3.3.1.4 Wire terminations.

3.3.1.4.1 Wire end connections. The ends of each wire, except for pigtail leads or parts requiring solder connections, shall be terminated by solderless crimp-type lugs for connections to screw-type terminals on parts or terminal boards. Parts with solder terminals or pigtail leads may have soldered connections. No more than three wires, including wires from parts, shall terminate at any one stud or terminal, except on multisection turret-type terminals where a maximum total of five wires may be terminated.

3.3.1.4.2 Cable connections. Cable connections between assemblies or subassemblies shall use connectors as specified herein.

3.3.1.4.3 Spare terminals. Terminal boards or cable connectors used to connect subassemblies within the control system shall have at least 10 percent spare terminals. There shall not be less than two such terminals. Spare terminals in connectors shall be in the outermost row of terminals. Where connectors or terminal boards are used only for primary power connections, no spare terminals need be provided. If more than one terminal board or connector is needed at a common location, only 10 percent of the total number of terminals at this location are required as spare terminals.

3.3.1.5 Wire identification. Internal wires of the control system shall be marked for identification at both ends, except for wires less than 300 mm (12 inches) in length, wires connecting terminals on a single part (unless such wires are laced or secured by sleeving into a wire bundle), and the end of wires terminating at connector terminals. Wires from 80 to 300 mm (3 to 12 inches) in length may be marked in the center only. Designations shall be marked on sleeve-type wire markers over the wire insulation. Markings shall be applied in a permanent manner, resistant to water, oil, and abrasion. Designations at each end shall consist of the part and terminal number, as assigned in the schematic diagram, for the connection at the end being marked, and the part and terminal number for the connection at the other end of the wire. Adhesive strip-type markers designed for wrapping around wires shall not be used. Wire markings shall be clearly visible in the assembled equipment.

3.3.1.6 Mounting of parts with pigtail leads (not PC boards). Parts with pigtail leads shall be secured between solder-type terminals on part mounting boards by soldering. The clearance between the soldered connection and the body of the part shall be not less than 4.8 mm (3/16 inch). Parts with pigtail leads whose mass exceeds 7 grams (0.0005 slugs) shall be held in place by means of body clamps to prevent damage or lead fatigue from shock and vibration. The pigtail leads of discrete semiconductors shall be covered by insulation sleeving, unless the solder terminals are located so that the leads cannot be shorted together or to other parts. The arrangement of parts shall be such that replacement of any part is possible without the removal of, or damage to, adjacent parts.

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3.3.1.7 Electrical bonding. Protective finishes shall be omitted at those points where their presence would prevent proper electrical bonding, as required for shielding or connections. Bonding shall be in accordance with MIL-STD-1310.

3.3.1.8 Grounding. Construction of the control system shall provide that exposed parts or panels of metal or other electrically conductive material are at ground (ship hull) potential at all times. In addition, the construction shall be such that grounding is not required for satisfactory operation. The control system shall not impose a ground upon the electrical power system from which it is energized.

3.3.1.9 Electrical protection. Protective devices, such as fuses or circuit breakers, shall be provided for protection of the control system from damage resulting from faults. These devices shall be inserted in both wires of single-phase, ac or dc circuits and in all three wires of 3-phase ac circuits. Protective devices shall be easily accessible for maintenance purposes.

3.3.1.10 Embedded parts. Parts shall not be embedded, except that parts used for temperature compensation may be embedded with the component being compensated.

3.3.1.11 Electrical insulation. Insulating materials shall conform to requirement 11 of MIL-STD-454.

3.3.1.11.1 Electrical creepage and clearance distances. The shortest point-to-point distance in air between energized parts or between an energized part and ground shall be not less than 1.6 mm (1/16 inch). The shortest point-to-point distance along the surface of an insulating material between energized parts or between an energized part and ground shall be not less than 1.6 mm (1/16 inch). These requirements do not apply to printed wiring boards.

3.3.1.11.2 Insulation resistance. The insulation resistance of completed control systems shall be not less than 10 megohms at 25 °C.

3.3.1.11.3 Dielectric strength. Completed control system assemblies shall withstand, without damage, a dielectric test voltage of 500 V. This requirement does not apply to printed wiring boards.

3.3.1.12 Equipment shielding and filtering. All instrumentation and control equipment incorporating active circuitry (such as, transistors and diodes) shall be provided with radio frequency (RF) shields, either separately or collectively. Such shields shall, as a minimum, consist of conducting enclosures, designed to provide at least 40 dB of shielding between 2 and 30 megahertz (MHz). No unprotected opening shall exceed 0.001 wavelength in maximum dimension at frequencies between 2 and 30 MHz. Conducting material not bonded to the edges of unprotected openings (such as, cables, components, chassis) shall be spaced away from such openings by at least twice the maximum dimension of the openings. Larger openings shall be protected with honeycomb filters; conducting glass; or conducting sleeves, having length at least twice the maximum cross-sectional dimension of the opening (waveguides below cutoff) bonded to the edges of the openings. Joints shall be either RF gasketed or welded. Conducting non-electrical penetrations (such as, bolts, pipes, and ducts) shall be bonded to the enclosure at the points or penetration and, if hollow, shall have a length at least twice the maximum cross-sectional dimension. Conducting electrical penetrations (such as, wires and

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cables) shall be filtered using electro-magnetic interference (EMI) filters having not less than 40 dB attenuation between 2 and 30 MHz when measured in accordance with MIL-STD-220, except that signals with information content above 500 kilohertz (kHz) shall not be filtered.

Each signal connected to the steering control system shall be low pass filtered inside the unit to attenuate frequencies not of interest to the control system. Unfiltered signals shall be run in shielded cables.

3.3.1.13 Cable separation. Cable separation within units shall be maintained in accordance with NAVSEA S9407-AB-HBK-010. Instrumentation and control cables shall be designated S4-7 for determining the minimum separation required.

3.3.2 Mechanical.

3.3.2.1 General construction. The control system shall be constructed of either steel or aluminum and shall be of the lightest weight consistent with the general and specific design requirements herein. The equipment shall be of fabricated construction.

3.3.2.2 Accessibility. Parts and subassemblies which may require servicing, repair, or replacement during the life of the equipment shall be readily accessible for repair or replacement.

3.3.2.3 Enclosures. The equipment enclosures shall conform to requirement 55 of MIL-STD-454 and as specified herein.

3.3.2.3.1 Degree of enclosure. The degree of enclosure for the control system shall be drip-proof (15 degrees) in accordance with MIL-STD-108.

3.3.2.3.2 Extraneous light. Enclosures shall prevent extraneous light escape.

3.3.2.3.3 Structural integrity. Self-resonances of the enclosure shall be greater than 50 Hz, when subjected to the vibration test of 4.6.19.

3.3.2.3.4 Rigidity. Flat-sheet metal surfaces, including protective covers, grills, and screens shall be of rigid construction, inflexible to the firm touch.

3.3.2.3.5 Rounded corners and edges. Edges and corners of surfaces normally exposed or painted shall be rounded. Sharp edges and points of any kind shall be avoided.

3.3.2.3.6 Moisture pockets. Moisture pockets within equipment enclosures shall be avoided. The design shall conform to requirement 31 of MIL-STD-454.

3.3.2.3.7 Handling. Handles shall be provided for removing units or assemblies from enclosures. Handles on the front panel of the equipment shall be positioned to protect the front panel indicators and controls when the units or assemblies are withdrawn and when the front panel is placed in a down position.

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3.3.2.3.8 External connections. When rack and panel techniques are not employed, the external connections to equipment enclosures shall be MS-type connectors. These connectors shall not be located on the front panel.

3.3.2.4 Mounting (panel). Equipment intended for front panel mounting shall incorporate a flange for securing the panel in a vertical position in the enclosure. The enclosure shall not project more than 25 mm (1 inch) from the face of the panel (not including operating handles).

3.3.2.5 Drilled and tapped holes. Drilled and tapped holes shall be slightly countersunk.

3.3.2.6 Controls, illumination, panel layout.

3.3.2.6.1 Controls. General criteria for characteristics of controls shall be in accordance with requirement 28 of MIL-STD-454, except that coupling between or to shafts shall be accomplished by means of rigid metallic or insulated couplings secured by pins.

3.3.2.6.2 Illumination. Controls, switches, and dials shall be illuminated by lighting sources integral with associated equipment. Light sources in the Bridge Steering Unit, including indicator lights, shall be dimmer controlled, either singly or in a group, either optically or electrically, from minimum discernible intensity to full brilliance, and shall provide uniform illumination. Dials and other displays shall be readable in all levels of illumination from 28 footcandles (white) to 0.03 footcandles (red) of incident illumination.

3.3.2.6.3 Dial lamps. Dial lamps shall not be connected in series. Lamp intensity for dial lamps in the Bridge Steering Unit shall be controlled from maximum value to minimum discernible intensity when all lamps are operative.

3.3.2.6.4 Design for dark-adapted areas. The indicating instruments on the ship control console shall appear white against a black background when viewed during the daylight and red when viewed during darkened conditions.

3.3.2.6.5 Dials and pointers for equipment having self-contained red illumination. Dials and pointers for units having self-contained red illumination shall have dark faces and white numerals, graduations, and lettering when viewed under high-level ambient illumination. They shall present red numerals, graduations, and lettering when the internal illumination is energized and viewed under low-level ambient illumination.

3.3.2.6.6 Dials and pointers for equipment not having self-contained red illumination. Dials and pointers for units not having self-contained illumination shall have white faces with black numerals, graduations, and lettering.

3.3.2.6.7 Dials and pointers. Dials and pointers shall conform to the following requirements. Dials shall be of plastic material. Pointers shall be of plastic material for illuminated units and of aluminum for nonilluminated units. The face of the dial with markings, lettering, and graduations shall be placed as close as practicable to the dial window. The dial faces, markings, lettering, graduations,

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and pointers shall have a matte finish. The markings shall be free from distortion, with clear and sharp edges. The pointer shall not cover the graduations to which it refers, but shall extend only to the nearer edge of the graduations.

3.3.2.7 Magnetic characteristics. The control system shall conform to the magnetic material, eddy current magnetism, and stray magnetic field requirements specified herein.

3.3.2.7.1 Magnetic materials. Except for material required to be magnetic for the electrical functioning of the system, the use of material which has a magnetic permeability of greater than 2.0 after fabrication shall be kept to a minimum. Such material shall be clearly identified on assembly drawings.

3.3.2.8 Special tools. The design and construction of the control system shall be so that the need for special tools for tuning, adjustments, maintenance, replacement, and installation is kept to a minimum. Special tools shall withstand the intended use throughout the life of the equipment.

3.3.3 Thermal.

3.3.3.1 Heat removal methods. The heat removal method selected shall be the simplest and most effective means that will ensure that components are maintained within their rated temperature limits under the environmental conditions specified.

3.3.3.1.1 Forced-air cooling. When forced air is used as a cooling method, the inlet-to-outlet temperature differential shall be not greater than 14 °C.

3.3.3.1.2 Water cooling. Water cooling shall not be used in the control system.

3.3.3.1.3 Ventilation openings. Ventilation openings shall be consistent with the degree of enclosure specified herein for each unit (see 3.3.2.3.1).

3.3.3.1.4 Air filters. When forced-air cooling is used, air filters shall be located at all air intakes and shall be readily removable for cleaning without disassembly of the equipment.

3.3.3.1.5 Cooling during maintenance. Equipment containing heat-critical parts, which may be damaged by excessive heat when a chassis is withdrawn for maintenance and adjustment, shall be provided with an integral cooling method to prevent such damage.

3.3.3.2 Parts application. Heat-sensitive parts shall be thermally isolated from heat sources by one or more of the following techniques.

a. Physical separation

b. Use of thermal insulation techniques

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c. Placement of heat-sensitive parts upstream in the cooling fluid or path, and heat sources downstream

d. Maximum use of conduction techniques to remove heat from heat dissipating parts along short metallic paths of large cross-sectional area to the heat sink or exchanger.

3.3.3.3 Surface temperature. Under any mode of normal operation, exposed parts, including the enclosure of the equipment, shall not exceed a temperature of 60 °C at an ambient temperature of 25 °C. The temperature of front panels and operating controls shall not exceed 49 °C at the same ambient temperature.

3.3.3.4 Thermal performance. The construction of the control system shall ensure that heat-sensitive parts are within the thermal stress identified in the reliability prediction report and any part derating requirement, and that the equipment conforms to thermal design requirements specified herein.

3.3.4 Parts, materials, and processes.

3.3.4.1 Parts. The control system shall use only those standard parts specified herein. The use of standard parts does not relieve the contractor of the responsibility for conforming to the performance requirements specified herein.

3.3.4.1.1 Selection of standard parts. The selection of standard parts shall be in accordance with the following order:

a. Standards, specifications, and requirements listed herein
b. Other standards, specifications, and requirements listed in MIL-STD-242.

3.3.4.1.2 Nonstandard parts. When a contractor has determined that the control system requirements cannot be met by using a standard part, the contractor shall determine the nonstandard part which will be suitable for the application and request written approval from the contracting activity. The requirements for the approval of nonstandard parts shall be in accordance with MIL-STD-965. Approval will not be granted for the use of parts of special or novel design where parts specified herein are suitable and available.

3.3.4.1.3 Individual selection. The performance of the control system shall not be dependent on the selection of individual parts for characteristics which have tolerances closer than those specified by the applicable part specification or military standard.

3.3.4.1.4 Tolerances. In the selection of parts, the widest tolerances permitted by the individual part specification shall be used in the design of the control system, commensurate with the particular long-term stable circuit application requirement. However, in the fabrication of the control system, parts of closer tolerances may be substituted as long as interchangeability is not affected.

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3.3.4.1.5 Plug-in parts. Plug-in parts shall be securely retained in their sockets in their proper position by means of positive holding devices to meet vibration and shock requirements. These parts shall also comply with the creepage and clearance requirements of 3.3.1.11.1.

3.3.4.1.6 Audible alarms. Audible alarms shall be selected and applied in accordance with MIL-A-15303 for drip-proof degree of enclosure.

3.3.4.1.7 Batteries. Batteries shall not be used in the control system.

3.3.4.1.8 Internal hook-up wire. Internal hook-up wire shall be selected and applied in accordance with requirement 20 of MIL-STD-454.

3.3.4.1.9 Capacitors. Except as specified herein, capacitors shall be selected and applied in accordance with MIL-STD-198. Paper dielectric capacitors, metallized paper capacitors, wet slug tantalum capacitors, and compression-type variable capacitors shall not be used. Fixed-dry electrolytic (aluminum oxide) capacitors shall conform to MIL-C-39018, and their use shall be restricted to power filter applications.

3.3.4.1.10 Circuit breakers. Circuit breakers shall conform to requirement 37 of MIL-STD-454. Only breakers conforming to MIL-C-17588 and MIL-C-17361 shall be used.

3.3.4.1.11 Connectors. Except as specified herein, connectors shall be selected and applied in accordance with MIL-STD-1353. Externally mounted connectors shall be provided with a suitable protective cap to prevent damage to the connector prior to installation.

3.3.4.1.11.1 Removable contact, crimp-type environmental connectors. These connectors shall have the full complement of contacts installed. Non-environmental-type connectors need only have the necessary contacts required by the equipment design and spares. Spare contacts shall meet requirement 10 of MIL-STD-454.

3.3.4.1.11.2 Mounting. Connectors shall be secured only by machine screws and shall be capable of ready removal and replacement.

3.3.4.1.12 Fuses and fuse holders. Except as specified herein, fuses and fuse holders shall be selected and applied in accordance with requirement 39 of MIL-STD-454.

3.3.4.1.13 Indicator lamps. Except as specified herein, indicator lamps shall conform to requirement 50 of MIL-STD-454. The color of the indicator lamps shall be in accordance with 3.3.2.6.4.

3.3.4.1.14 Meters. Meters used for electrical indication, elapsed time, and frequency indication shall conform to requirement 51 of MIL-STD-454.

3.3.4.1.15 Digital indicators. Digital indicators shall be as specified for each unit in the paragraphs on performance characteristics.

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3.3.4.1.16 Jacks (test point). Tip jacks shall be used for test point jacks and shall be selected in accordance with MIL-STD-242. Protection shall be provided in test point circuitry to prevent damage to the equipment caused by accidental grounding of test points. Banana jacks shall not be used as test points.

3.3.4.1.17 Relays. Except as specified herein, relays shall be selected and applied in accordance with MIL-STD-1346. Solid-state relays shall be selected in accordance with MIL-STD-1346 and conform to MIL-R-28750.

3.3.4.1.18 Resistors. Except as specified herein, resistors shall be selected and applied in accordance with MIL-STD-199. Variable wire-wound resistors of less than 10-watt rating shall not be used, unless they are of the precision type.

3.3.4.1.19 Locking devices on resistors. Variable resistors mounted on chassis or panels shall have positive locking devices. Locking devices shall not be required on miniature potentiometers that meet the stability requirements of requirement 28 of MIL-STD-454.

3.3.4.1.20 Semiconductor devices. Semiconductor devices shall be in accordance with requirement 30 of MIL-STD-454.

3.3.4.1.21 Microcircuits. Microcircuit technology shall be considered in the design of control systems, and the maximum use shall be made of standard microcircuits. The selection and application of microcircuits and microelectronic devices shall conform to requirement 64 of MIL-STD-454.

3.3.4.1.22 Switches. Switches and associated hardware shall be specified and applied in accordance with MIL-STD-1132. Required switches that are not listed in MIL-STD-1132 shall conform to one of the following specifications: MIL-S-12285, MIL-S-15743, MIL-S-18396, MIL-S-21604, MIL-S-83731.

3.3.4.1.23 Synchros. Synchros shall be in accordance with MIL-S-20708. Installation requirements shall be in accordance with MIL-HDBK-225.

3.3.4.1.24 Terminal boards. Terminal boards shall be in accordance with requirement 19 of MIL-STD-454.

3.3.4.1.25 Thermistors. Temperature-compensating thermistors shall be in accordance with MIL-T-23648.

3.3.4.1.26 Transformers and inductors. Transformers, inductors, and coils shall be in accordance with requirement 14 of MIL-STD-454.

3.3.4.1.27 Lacing cord and tape. Lacing cord shall be in accordance with type P of MIL-T-713. Lacing tape shall conform to type I, size 3, finish B of MIL-T-43435.

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3.3.4.1.28 Sleeving. Sleeving shall be synthetic resin, type A or F, class I, category 1, in accordance with MIL-I-631. Heat-shrinkable-type sleeving shall be in accordance with MIL-I-23053.

3.3.4.1.29 Cable straps and clamps. Cable clamps and straps shall be used only for tiedown applications and shall be selected in accordance with MIL-STD-242.

3.3.4.1.30 Air filters. Air filters for ventilating systems of the control system shall be of the cleanable type. These air filters shall be mounted so as to be capable of being readily examined and replaced from the outside of the equipment, when the equipment is normally installed. Filters shall conform to MIL-F-16552.

3.3.4.1.31 Handles and control knobs. Handles and control knobs shall be selected and applied in accordance with MIL-STD-242.

3.3.4.1.32 Fastening devices. Fastening devices shall be selected and applied in accordance with requirement 12 of MIL-STD-454, except that threaded fasteners shall be selected from MIL-STD-1251 whenever possible.

3.3.4.1.33 Standard electronic modules (SEMs). SEMs shall be selected and applied in accordance with requirement 73 of MIL-STD-454.

3.3.4.1.34 Printed wiring. Printed wiring shall be in accordance with requirement 17 of MIL-STD-454.

3.3.4.2 Materials.

3.3.4.2.1 General. The materials used in the construction of the control system shall be of a type, class, form, and grade which is readily available from normal sources of supply, without the necessity for additional treatment or processing other than that which is normal to, or readily supplied by, the industry. Such materials are standard materials. Other, or special materials, may be used only when adequately justified, taking into account both the technical and economical aspects and considering maintenance and support requirements, as well as initial supply. The use of these special materials shall require approval from the contracting activity.

3.3.4.2.2 Prohibited materials. Materials which are capable of producing dangerous volumes of gases or other harmful effects under high temperature conditions, including fire, shall not be used. Materials which will ignite or explode from an electric spark, flame, or from heating, and which if so ignited will independently support combustion in the presence of air, shall not be used. Any material which is fragile in the form, size, or manner in which it would be used shall not be used. Materials in a frail form, which are not positively protected against mechanical damage as used in a part or subassembly, fall within the prohibited fragile category. Mercury, magnesium, magnesium-base alloys, cadmium, and radioactive materials shall not be used in the construction of the control system.

3.3.4.2.3 Arc-resistant materials. Arc-resistant materials shall conform to requirement 26 of MIL-STD-454.

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3.3.4.2.4 Metals. Metals and alloys shall be corrosion resisting or shall be given a corrosion-resistant treatment or coating.

3.3.4.2.5 Plastics. Plastics that are used primarily for electrical insulation shall be as specified in 3.3.1.11. Plastics that are used as structural parts shall meet the physical and mechanical properties required for plastic insulating materials, including nonflammability and nontoxicity; however, these plastics need not meet the arcing- and tracking-resistance requirements.

3.3.4.2.6 Ceramics. Ceramic materials shall not be used for insulating purposes without approval from the contracting activity.

3.3.4.2.7 Encapsulation and embedment. The encapsulation or embedment of assemblies shall conform to requirement 47 of MIL-STD-454. Rigid materials used for embedding electronic parts in assemblies shall conform to MIL-I-16923.

3.3.4.2.8 Dissimilar metals. Metal-to-metal contact shall be limited to those metals which, when coupled, will have minimal corrosion in the vicinity of the contact.

3.3.4.3 Processes.

3.3.4.3.1 Welding. Structural welding shall conform to requirement 13 of MIL-STD-454.

3.3.4.3.2 Soldering. Soldering shall conform to requirement 5 of MIL-STD-454.

3.3.4.3.3 Brazing. Brazing of steel, copper, copper alloys, nickel alloys, aluminum, and aluminum alloys shall be in accordance with MIL-B-7883.

3.3.4.3.4 Treatment and processing of metals for corrosion resistance. Corrosion-resistant treatment shall not be applied to surfaces which are in functional contact, where gouging or binding may be a factor, where the treatment might interfere with normal functioning, or where electrical grounding through the surface is required. Corrosion protection and corrosion-resisting treatments shall conform to requirement 15 of MIL-STD-454 and as specified herein.

3.3.4.3.5 Painting. The exterior and interior surfaces of metal equipment enclosures shall be painted as specified herein. Prior to painting, the applicable pretreatment and primer shall have been completed. Plastic enclosures shall not be painted.

3.3.4.3.5.1 Aluminum and aluminum alloy pretreatment. Aluminum and aluminum alloy pretreatment shall be as follows:

a. *Cleaning.* The basic metal shall be cleaned to remove grease, oil, welding flux, and other foreign matter.

b. *Application.*

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(1) *Protected equipment.* For protected equipment (interior use), aluminum and aluminum alloys shall be anodized in accordance with MIL-A-8625, chemically treated in accordance with MIL-C-5541, or pretreated in accordance with type III of TT-C-490.

(2) *Exposed equipment.* For exposed equipment, aluminum and aluminum alloys shall be chemically treated in accordance with MIL-C-5541.

3.3.4.3.5.2 Ferrous metal treatment. Ferrous metal treatment shall be as follows:

a. *Cleaning.* After machining, welding, and brazing operations are completed, rust and other corrosive products and flux shall be removed by abrasive blasting, sanding, wire brushing, or other mechanical means. Surfaces shall be cleaned of all grease, oil, and dirt by solvent wiping and rinsing, vapor degreasing, or caustic washing followed by rinsing.

b. *Application.* Ferrous metals shall be pretreated in accordance with type I or III of TT-C-490.

3.3.4.3.5.3 Primer. Primer shall be applied in accordance with one of the following methods:

a. One coat of primer pretreatment in accordance with DOD-P-15328 0.005 to 0.013 mm dry film thickness (0.0002 to 0.0005 inch) followed by primer conforming to TT-P-1757 or TT-P-664.

b. A hot-dip tank phosphate treatment conforming to type I of TT-C-490 0.015 to 0.025 mm dry film thickness (0.0006 to 0.001 inch) followed by primer conforming to TT-P-1757 or TT-P-664.

c. Primer conforming to TT-P-1757 or TT-P-664 shall be applied as one coat 0.015 to 0.020 mm dry film thickness (0.0006 to 0.0008 inch).

3.3.4.3.5.4 Enamel. Enclosures shall be painted with two continuous film coats of gray enamel conforming to MIL-E-15090. Each coat shall have a minimum thickness of 0.025 mm (0.001 inch).

3.3.5 Electromagnetic radiation. Major components of the steering control system shall meet the emission and susceptibility requirements of parts 1 and 5 of MIL-STD-461. Acceptance testing shall be in accordance with MIL-STD-462. Applicable requirements are listed in table I.

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TABLE I. *Applicable EMI requirements.*

Requirement	Description
CEO1	Conducted emission, power and interconnecting leads, low frequency (up to 15 kHz)
CEO3	Conducted emissions, power and interconnecting leads (0.015 to 50 MHz)
CSO1	Conducted susceptibility, power leads, 30 Hz to 50 kHz
CSO2	Conducted susceptibility, power leads, 0.05 to 400 MHz
CSO6	Conducted susceptibility, power leads, spikes
REO1	Radiated emissions, magnetic field, 0.03 to 50 kHz
REO2	Radiated emissions, magnetic field, 14 kHz to 10 gigahertz (GHz)
RSO1	Radiated susceptibility, magnetic field, 0.03 to 50 kHz
RSO2	Radiated susceptibility, magnetic induction field, spikes and power frequencies
RSO3	Radiated susceptibility, electric field, 14 kHz to 40 GHz

3.3.6 Identification and marking.

3.3.6.1 General. Identification and information plates shall be provided for each panel and local control unit in the steering control subsystem (see 3.2.2.1 through 3.2.2.3). The plates shall be attached by means of screws in tapped holes, screws held by captive nuts, or screws held by nuts welded to the opposite side of the panel or chassis. The mounting and location of the plates shall not adversely affect the strength of the item on which the plates are mounted or impair the integrity of the enclosure. Aluminum plates shall be attached by corrosion-resisting screws and not by aluminum screws. Plastic information plates shall be mounted by corrosion-resisting screws and shall be gray in color with lettering in black.

3.3.6.2 Identification plates. Identification plates shall be metallic and in accordance with types A, B, C, or H of MIL-P-15024. The following minimum information shall be provided and may be the manufacturer's usual identification plate. Identification plates shall conform to MIL-P-15024 except for size.

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a. Name of panel or unit
b. Manufacturer's name and Commercial and Government Entity (CAGE) code
c. Contract number
d. Part number
e. Serial number
f. Technical manual number (if available)
g. Salient design characteristics (the pertinent rating, operating characteristics and other information necessary for identification of the item).

3.3.6.3 Information plates. Information plates shall be in accordance with types A, B (metallic only), C, or H of MIL-P-15024.

3.3.6.3.1 Warning and caution plates. Plates showing safety notices or operating instructions which, if not observed, would result in material damage or personal injury, shall have the word WARNING or CAUTION in double-sized letters filled with paint the color selected in accordance with ANSI Z53.1. The use of warning and caution plates shall be limited to the following conditions:

a. *WARNING* – Where nonobservance would result in personal injury, loss of life, or damage to the equipment

b. *CAUTION* – Where nonobservance would result in damage to, or destruction of, equipment or inadvertent shutdown of a critical system.

3.3.6.4 Identification of parts by reference designation. To facilitate maintenance, each part shall be identified by its reference designator as assigned in the schematic diagram.

3.3.6.4.1 Reference designation location. Reference designators shall be located adjacent to each part and shall be marked on the chassis, back or front panel, partitions, insulator strips, or any surfaces which are not subject to replacement. Designations shall be marked in such a position as to physically locate the parts and still be readily visible for purposes of maintenance without removal of other parts.

3.3.6.4.2 Enclosed parts. Reference designators for parts enclosed in separate and removable shields or compartments may be marked on the shields or supporting structures for such parts, provided that the replacement of such parts does not require destruction of the original shield or supporting structures, and provided that such shields or supporting structures are not interchangeable with other shields or structures in the control system.

3.3.6.4.3 Identification of part leads. For each polarized part, such as electrolytic capacitors and rectifiers, the polarity identification shall be marked on the part-mounting surface in order to ensure proper replacement of the part. For multiple lead parts, such as transistors or microcircuits, where proper circuit operation is dependent on a specific lead connection, appropriate lead identification shall be marked on the part-mounting board to ensure proper replacement of the part.

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3.3.6.4.4 Method of marking. Markings shall be permanent and legible. The markings on plastic or metallic materials shall be made by stamping, engraving, stenciling, rubber stamping with smudge-proof ink covered with a coat of clear lacquer, or silk screening. Decals or paper labels shall not be used.

3.3.7 Workmanship. Workmanship shall conform to requirement 9 of MIL-STD-454.

3.3.8 Interchangeability. Parts, panels, subassemblies, and assemblies which are removable shall be interchangeable with corresponding items from similar equipment or from production, without the necessity for reforming (for example, drilling, bending, or filing), without the use of undue force, and without the necessity of conditional selection.

3.3.9 Safety.

3.3.9.1 Electrical. The steering control system shall incorporate methods to protect personnel from accidental contact with voltages in excess of 30 Vac (rms) or dc during normal operation of a complete equipment. Means shall be provided so that ac power may be cut off while installing, replacing, or interchanging a complete equipment, assembly, or parts thereof.

3.3.9.2 Grounding. Ground connection to an electrically conductive chassis or frame shall be mechanically secured by soldering to a spot-welded terminal lug or to a portion of the chassis or frame that has been formed into a soldering lug, or by using a terminal on the ground wire and securing the terminal by a screw, nut, and lockwasher. The screw shall fit in a tapped hole in the chassis or frame or it shall be held in a through-hole by a nut. When the chassis or frame is made of steel, the metal around the screw hole shall be plated or tinned to provide a corrosion-resistant connection. When aluminum or aluminum alloys are used, the metal around the grounding screw or bolt hole may be covered with a corrosion resistant surface film only if the resistance through the film is not more than 0.002 ohm.

3.3.9.3 Mechanical. The steering control system shall provide maximum access and safety to personnel while installing, operating, and maintaining the equipment. Sharp projections on cabinets, doors, and similar parts shall be avoided.

3.3.9.4 Interconnections. The steering control system shall provide means to prevent inadvertent reversing or mismatching of connectors.

3.3.9.5 Marking. Guards, barriers, access doors, covers, or plates shall be marked to indicate the hazard which may be reached upon removal of such devices. Hazards exposed upon removal of such devices, and significantly different from those of surrounding items, shall be marked adjacent to the hazards.

3.3.10 Human performance/human engineering. Human engineering design criteria and principles used in the design of the control system shall be in accordance with ASTM F-1166.

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3.3.11 Use of standard electronic modules. Standard electronic modules (SEMs) shall be used in the steering control system (see 6.3). SEMs shall be acquired in accordance with MIL-M-28787 for class II modules and implemented in accordance with MIL-STD-1378. The use of special modules shall be minimized and, where implemented, shall conform to the design requirements of MIL-STD-1389 and class II environmental requirements of MIL-M-28787. All special modules shall be inspected in accordance with the inspection conditions and qualification inspection requirements (excluding correlation samples, retention of qualification, and sequence of testing requirements) of MIL-M-28787.

3.4 System safety program. A safety assessment shall be performed (see 6.3). Safety design features, including fail-safe features, shall be incorporated into the design to prevent damage to equipment and to ensure optimal personnel protection during installation, operation, maintenance, and repair or interchanging of any component or assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Installation of test item in test facility. The test item shall be installed in the test facility in a manner that will simulate service usage, making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test item is installed. The test item shall then be tested to determine that no malfunction or damage was caused due to faulty installation procedures or handling. The requirement for operating following installation of the test item in the test facility is applicable only when operation is required during or immediately following exposure to the specific test.

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4.1.3 Pre-performance test. Prior to proceeding with any of the test methods, the test item shall be operated under standard ambient conditions (see 4.1.6), and information necessary to determine compliance with required performance taken. This information shall provide the criteria for checking satisfactory performance of the test item either during or at the conclusion of the test or both, as required.

4.1.4 Performance check during tests. When operation of the test item is required during the test exposure, the operation and performance checks shall be of sufficient duration and shall be repeated at appropriate intervals to ensure a record of comprehensive comparative information for comparison with information recorded under standard ambient conditions. No adjustments, alignment, or realignment of equipment shall be permitted during and after environ-mental tests. No repairs shall be permitted after exposure of equipment to extreme ambient conditions before making the required measurements. No alignment, realignment, adjustments, or repairs shall be undertaken prior to completion of all tests unless approved by the contracting activity.

4.1.5 Visual examination and failure criteria. When specified herein, the test item shall be visually examined and a record made of any damage or deterioration resulting from the test. If a test chamber is used for the test, a visual examination of the test item within the chamber shall be performed during the test, when possible. Upon completion of the test, a visual inspection of the test item shall be performed again, after the test item has been returned to standard ambient conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service or maintenance requirements shall provide reason to consider the test item as having failed to withstand the conditions of the test. When a failure or defect is detected, the cause shall be determined and corrective measures ascertained. These corrective measures shall be introduced into new production and shall also be applied to completed units, unless allowed otherwise by the contracting activity.

4.1.6 Test conditions. Tests shall be conducted with the test item operating under the following specified conditions, except where these conditions are the test variables:

a. The ambient temperature shall be 25 $\pm$ 5 °C.

b. The relative humidity shall be between 25 and 50 percent.

c. AC supply voltage shall be the normal operating voltage.

d. AC supply frequency shall be the normal operating frequency.

e. Positions of the controls shall be in the neutral or normal operating position.

f. All electrical inputs shall be connected to a signal source having parameters similar to those expected in normal operation, or connected to signal ground.

g. All electrical outputs shall be connected to a load having parameters similar to those expected in normal operation. Outputs classified as signal outputs can be left unconnected.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspections (see 4.4).

4.3 First article inspection. Unless otherwise specified (see 6.2 and 6.3), the first article inspection required shall be as specified in table II. The contractor shall conduct any other inspection required to determine compliance with the performance requirements specified herein.

TABLE II. *Examinations and tests.*

Examination or test	Requirement	First article inspection	Quality conformance inspection
General examination	3.3	4.5	4.5
Dimensions and weights	3.2.4, 3.2.3.3.1.1, 3.2.3.3.2.1, 3.2.3.3.3.1, 3.2.3.3.7.1, and 3.2.3.3.8.1	4.5.1	–
Unit performance		4.6.1	4.6.1
Integrated performance		4.6.2	–
Dielectric strength	3.3.1.11.3	4.6.3	4.6.3
Insulation resistance	3.3.1.11.2	4.6.4	4.6.4
Adjustable controls setting verification	3.2.7	4.6.5	–
Supply line voltage and frequency	3.3.1.1	4.6.6	
Transient voltage and frequency	3.3.1.1.1	4.6.7	–
Power interruption	3.3.1.1	4.6.8	–
Part interchangeability	3.3.4.1 and 3.3.8	4.6.9	– –
Enclosure	3.3.2.3	4.6.10	–
Humidity	3.2.7.2	4.6.11	–
Low temperature	3.2.7.1.1	4.6.12	–
High temperature	3.2.7.1.2	4.6.13	–
Electromagnetic interference emission and susceptibility	3.3.5	4.6.14	–
Airborne and structureborne noise	3.2.7.5	4.6.15	–
Roll and pitch	3.2.7.6	4.6.16	–
Inclination	3.2.7.7	4.6.17	–
Overvoltage	3.3.1.1.2	4.6.18	–
Vibration	3.2.7.3	4.6.19	–
High impact shock	3.4.7.4	4.6.20	–
Power (ac)	3.3.1.1	4.6.21	–
Stress screening		4.6.22	4.6.22
Automatic steering control	3.2.3.2.10.6	4.6.23	–

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4.4 Quality conformance inspection. Quality conformance inspection shall be conducted on each major unit of the control system prior to delivery (see 6.3). This inspection shall consist of the examinations and tests specified in table II.

4.4.1 Test program. A test program shall be developed for testing and stress screening each steering control unit as follows:

a. Testing the performance as a unit
b. Testing the performance as an integral part of a system configuration
c. Electrical/environmental stress screening of each unit.

4.4.1.1 Test plan. A test plan (see 6.3) shall provide for testing the performance of each steering control unit as a unit and as an integral part of a system configuration (see 4.6.1 and 4.6.2). The test plan shall also provide for electrical/environmental stress screening of each unit of the steering control system on a 100-percent basis, in accordance with NAVMAT P-9492 (see 4.6.22). The test plan shall define screening procedures at the unit level. The plan shall be dynamic in nature and structured so that maximum screening effectiveness is achieved with minimum cost. The types, manner of application, and level of severity of each electrical/environmental screen shall be defined. Unless more effective screens are determined through proofing experiments, the stress screening portion of the test plan shall include environments and levels as follows:

Thermal cycling:

Temperature range: -40° to +90 °C

Temperature rate of change: 5 °C per minute minimum, as measured by thermocouples mounted directly on electronic components

Thermal cycles: 10 – 20 cycles as a minimum, depending on level of assembly (last cycle failure free)

Operating conditions: Power ON only between specified lower and upper operating temperatures, except for modules on mature production hardware

Random vibration:

Duration: 10 minutes per axis

Number of axes: Two axes

Acceleration spectrum: 0.04 g^2/Hz, from 20 to 2,000 Hz with a 3-dB/octave rolloff from 80 to 20 Hz, and 350 to 2,000 Hz.

4.5 General examination.

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4.5.1 Dimensions and weights.

4.5.1.1 Uncrated. Dimensions and weights of all units comprising the complete control system and all the repair parts (not crated for shipment) shall be determined.

4.5.1.2 Crated. Dimensions and weights of all units comprising the control system and all the repair parts (crated for shipment) shall be determined. It shall be ascertained that a complete list of contents is included in each shipping container.

4.6 Tests.

4.6.1 Unit performance test. Each major unit of the system shall be subjected to a general performance test to demonstrate conformance with the requirements specified herein. Tests shall be in accordance with the test plan as specified in 4.4.1.1 (see 6.3).

4.6.2 Integrated performance test. All units, assemblies, and panels of the control system shall be interconnected in such a manner as to duplicate the ship configuration. The system shall then be subjected to a performance test to demonstrate conformance with the requirements specified herein. Tests shall be in accordance with the test plan as specified in 4.4.1.1 (see 6.3).

4.6.3 Dielectric strength. The completed control system shall be subjected to a dielectric strength test. For first article inspection, the equipment shall withstand a test voltage of 500 V for a period of not less than 1 minute. For quality conformance inspection, equipment shall withstand a test voltage of 500 V for 5 $\pm$ 1 seconds. The frequency of the test voltage shall not be less than 60 Hz and shall approximate a true sinewave. The source of the test potential shall have a rating of at least 1.0 kilowatt (kW). Assemblies having circuits grounded to the chassis shall have the ground to chassis disconnected before testing. Shielding shall not be disconnected from ground. Low voltage parts, such as transistors, diodes, micro-circuits, electrolytic capacitors, and other voltage-sensitive parts shall be disconnected during this test.

4.6.3.1 Dielectric strength test procedures. The test voltage shall be applied between electrically isolated circuits and between each circuit and ground. The test voltage shall be increased gradually to the specified value and held at that value for the specified time.

4.6.3.2 Dielectric strength failure criteria. Any audible or visible evidence of arcing, corona, or punctured insulation during or after the test shall be reason for rejection.

4.6.4 Insulation resistance. The completed control system shall be subjected to an insulation resistance test. The test voltage shall be applied for a period of not less than 60 seconds and shall be conducted with the equipment cold at any convenient room temperature. At the time of the test, the temperature of the circuits shall be measured and information taken. The insulation resistance measurements shall be corrected to 25 °C. These corrections shall be made on the basis of insulation resistance doubling for each 15 °C decrease in temperature. The relative humidity at the time of the test shall be measured and recorded. The grounds for any grounded circuit shall be disconnected, and ungrounded circuits shall be thoroughly discharged by momentarily shorting them to ground prior to

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testing. Shielding shall not be disconnected from ground. Low voltage parts such as transistors, diodes, microcircuits, electrolytic capacitors, and other voltage-sensitive parts shall be disconnected during this test.

4.6.4.1 Insulation resistance test procedures. The test voltage shall be applied between electrically isolated circuits and between each circuit and ground. Circuits may be tied together so that only one test voltage need be applied when testing between the circuits and ground, providing that the insulation resistance when tested in this manner meets the minimum specified value.

4.6.5 Adjustable controls setting verification. The deenergized control system shall be placed in an ambient temperature environment o 25 $\pm$ 5 °C for a period of not less than 8 hours. The control system shall then be energized for a period not greater than 30 minutes. The setting of controls and adjustments shall be as specified in the approved test procedures. The accuracy of the output and of each set point shall be determined immediately after this period. The accuracy shall be within the performance accuracies specified in 3.2.3.

4.6.6 Supply line voltage and frequency. The equipment shall be operated for a period of not less than 15 minutes for each supply line voltage and frequency condition shown in table III. Reference measurements shall be made at nominal voltage and frequency (test a). The equipment shall then be operated in the combinations of voltage and frequency shown in table III and the performance characteristics, such as the accuracy of each set point and of the output, shall be determined for each condition. The settings of controls and adjustments shall be as specified in the approved test procedures. The combinations of voltage and frequency tests shown in table III shall be conducted at 0 °C, +25 °C and +50 °C.

TABLE III. *Supply line voltage and frequency.*

Test	Voltage (Percent of nominal)	Frequency (Percent of nominal)
a	100	100
b	110	95
c	110	105
d	90	105
e	90	95

4.6.7 Transient voltage and frequency.

4.6.7.1 Transient voltage upper limit. With the equipment operating in the upper limit of steady-state voltage, a transient voltage of plus 18 percent of normal voltage, recovering to the steady-state voltage within 2 seconds, shall be superimposed. The equipment shall operate normally following the transient.

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4.6.7.2 Transient voltage lower limit. With the equipment operating on the lower limit of steady-state voltage, a transient voltage of minus 18 percent of normal voltage, recovering to the steady-state voltage within 2 seconds, shall be superimposed. The equipment shall operate normally following the transient.

4.6.7.3 Transient frequency upper limit. With the equipment operating at 5 percent above the normal frequency, the frequency shall be increased by an additional 3 percent, recovering to the steady-state frequency within 3 seconds. The equipment shall operate normally following the transient.

4.6.7.4 Transient frequency lower limit. With the equipment operating at 5 percent below the normal frequency, the frequency shall be decreased by an additional 3 percent, recovering to the steady-state frequency within 2 seconds. The equipment shall operate normally following the transient.

4.6.8 Power interruption. With the equipment operating within the steady-state tolerances of voltage and frequency, the external power supply shall be suddenly interrupted, and after an interval between 3 and 4 seconds, the power supply, within the steady-state tolerances, shall be reapplied. After the equipment has been operating long enough to detect any major performance degradation and to include any recycling time, the power shall be interrupted for an interval of 29 to 30 seconds. The cycle, consisting of three such interruptions, shall be repeated for a total of four times. During, and as a result of these tests, no damage to the equipment shall be incurred, and any noted effects of power interruption or reapplication shall fall within the performance tolerances specified herein. Power interruption tests shall be performed when the control system is in the Manual and Automatic modes, as well as the condition where only the local control units are operable.

4.6.9 Parts interchangeability. The part interchange test need be conducted on only one of each type of identical assembly in the control system. The approved test procedure shall designate each part to be interchanged. Each of these parts shall be removed with the equipment deenergized and replaced by one of the same type chosen at random from stock. The equipment shall then be energized, and the unit performance test (see 4.6.1) and supply line voltage and frequency test (see 4.6.6) shall be conducted. The test shall be repeated with another set of the same parts chosen at random from stock. The equipment may be realigned after each interchange and prior to the operating test, using only the adjustment controls in the assembly in which a part or parts were replaced.

4.6.9.1 Interchanged parts. The parts to be interchanged shall include the following:

a. Each individually selected part, if any
b. Each set of matched parts, if any
c. Each part whose tolerance is critical with regard to circuit operation.

4.6.10 Enclosure test. The enclosure test shall be conducted in accordance with MIL-STD-108, for drip-proof (15 degrees) protected enclosure.

4.6.11 Humidity tests. Humidity tests shall be in accordance with method 507, procedure IV, of MIL-STD-810.

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4.6.12 Low temperature test. The low temperature test shall be in accordance with method 502 of MIL-STD-810, except as follows:

a. Step 2. Storage or nonoperating temperature shall be −55 °C (+0, −5 °C), for a duration of 24 hours.

b. The combined duration of steps 4 and 5 shall be not less than 12 hours, at least two of which shall be in the stabilized condition prior to conducting tests.

4.6.13 High temperature tests. The high temperature tests shall be in accordance with method 501, procedure I, of MIL-STD-810, except as follows:

a. Applicable high operating temperature shall be 50 °C (+5, −0 °C).

b. The combined duration of steps 4 and 5 shall be not less than 12 hours, at least six of which shall be in the stabilized condition prior to conducting tests.

c. Step 7. The equipment shall operate for a period of at least 2 hours in the stabilized condition before step 7 tests are performed.

4.6.14 Electromagnetic interference emission and susceptibility tests. Electromagnetic interference emission and susceptibility tests shall be in accordance with the test methods of MIL-STD-462.

4.6.15 Airborne and structureborne noise. Airborne sound measurement tests shall be in accordance with MIL-STD-740-1 and structureborne noise tests shall be in accordance with MIL-STD-740-2. Equipment shall meet the requirements specified in 3.2.7.5.

4.6.16 Roll and pitch test. The control subsystem shall be mounted on any device capable of performing the test. The control subsystem shall be rolled at the rate of 5 to 7 cycles per minute in one plane, to angles of 45 degrees to either side of the vertical. The duration of the test shall be of sufficient length to determine the control subsystem characteristics under such motion or for of not less than 30 minutes. The test shall be repeated with the control subsystem re-oriented 90 degrees horizontally from the plane in which it was originally tested.

4.6.17 Inclination test. At the conclusion of the roll and pitch test of 4.6.16, the cyclic motion shall be stopped and the inclination adjusted to an angle of 15 degrees. The equipment shall then be operated for a sufficient period to ensure that continuous operation can be maintained. The test to verify continuous operation at 15 degree inclination shall be repeated for each 90 degree orientation about the vertical axis.

4.6.18 Overvoltage. The equipment shall be energized at 149 V at the temperature of +50 °C (+5, −0 °C) for a period of 12 continuous hours. Modules and protective covers shall be closed during this test. Temperature-measuring sensors shall be placed at critical points throughout the equipment covering suspected areas of high temperatures. Temperatures shall be taken every 3 hours

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and at the conclusion of the test. Near the end of the test, the equipment shall be demonstrated at being operable.

4.6.19 Vibration. Equipment shall be subjected to the type I vibration test in accordance with MIL-STD-167-1, from 4 Hz (or lowest attainable frequency) up to and including 50 Hz. At the conclusion of the vibration tests, the dielectric strength test specified in 4.6.3 shall be conducted, except that the test voltage shall be 325 V. The equipment shall be fully energized during vibration tests.

4.6.20 High impact shock. The equipment shall be energized and fully operating during and after shock tests. Simulated loads such as weights or other dummy masses shall not be used. The test shall be performed with the normal assemblies in their functionally operating positions. Shock tests shall be in accordance with test methods of MIL-S-901.

4.6.21 Power (ac). Measurement of input power shall be made at normal line voltage and frequency under each principal mode of operation of the equipment. Power input measurements shall be taken during these tests.

4.6.22 Stress screening. The tests performed while stress screening each unit shall be in accordance with the test plan as specified in 4.4.1.1 (see 6.3).

4.6.23 Automatic steering control test. Autopilot performance shall be tested by providing a hardware in-the-loop computer simulation. The simulation developed for the design of the autopilot shall be used or duplicated and shall provide inputs to the actual autopilot. The autopilot outputs shall be connected back to the computer simulation. The hardware-in-the-loop simulation test shall demonstrate that the fuel economy mode provides fuel saving in excess of 1.5 percent over a well-tuned proportional integral/derivative controller. The test shall simulate conditions present when operating at the design speed at random headings, in seaways characterized by a two-parameter spectrum Bretschneider (significant wave height, modal wave period) with a cosine squared wave energy spreading function. The significant wave height/modal wave period pairs shall be: 1.5 meters/7 seconds, 3.5 meters/9 seconds, 5.5 meters/9.5 seconds, and 7.5 meters/11 seconds. An Automatic Steering Control Panel shall be provided for the manual control of the autopilot during these simulation tests. The hardware-in-the-loop simulation test shall demonstrate an impulse test to excite resonances and uncover marginal stability conditions which may show up as lightly damped transient oscillations on one or more channels of the chart recordings. The simulation shall demonstrate an adaptive response test to verify the ability of the autopilot to rapidly adapt to changing ship headings in selected seaways at selected ship speeds as the ship turns.

4.7 Inspection of packaging. Sample packs and the inspection of the preservation, packing, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging. Preservation, packing and marking shall be in accordance with the individual equipment specification and the level(s) of packaging specified (see 6.2).

6. NOTES

6.1 Intended use. This specification is intended to incorporate those general requirements which are common to the procurement of steering control systems for all Navy surface ships.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. Whether first article inspection is required (see 3.1)

d. If compatibility with Data Multiplex System is required (see 3.2.3.2.8.2, 3.2.3.3.1.2.h., 3.2.3.3.2.2.j., 3.2.3.3.3.2.h., 3.2.3.3.5, 3.2.3.3.6, and 3.2.9.3)

e. When algorithms for roll stabilization are required (see 3.2.3.2.10.6)

f. If active rudder roll stabilization is required (see 3.2.3.3.2.2(l))

g. What software executives are to be used (see 3.2.10.1)

h. Whether first article inspection is other than as specified (see 4.3)

i. Levels of packaging required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference paragraph	DID number	DID title	Suggested tailoring
3.2.5	DI-R-7079	Reliability Program Plan	–
3.2.6	DI-MNTY-80822	Maintainability Program plan	–
3.2.10	DI-QCIC-80572	Software Quality Program Plan	–
3.2.10	DI-MCCR-80012	Software Design Document	–
3.2.10	DI-MCCR-80013	Version Description Description	–
3.2.10	DI-MCCR-80014	Software Test Plan	–
3.2.10	DI-MCCR-80015	Software Test Description	–
3.2.10	DI-MCCR-80017	Software Test Report	–
3.2.10	DI-MCCR-80018	Computer System Operator's Manual	–
3.2.10	DI-MCCR-80019	Software User's Manual	–
3.2.10	DI-MCCR-80024	Computer Resources Integrated Support Document	–
3.2.10	DI-MCCR-80025	Software Requirements Specification	–
3.2.10	DI-MCCR-80026	Interface Requirments Specification	
3.2.10	DI-MCCR-80027	Interface Design Document	–
3.2.10	DI-MCCR-80029	Software Product Specification	–

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Reference paragraph	DID number	DID title	Suggested tailoring
3.2.10	DI-MCCR-80030	Software Development Plan	
3.3 and appendix	DI-DRPR-80651	Engineering Drawing	Level 3
3.3.11	DI-E-7042	Design Approval Request for Standard Electronic Modules Program	–
3.3.11	DI-E-7044	Request, Exception, Standard Electronic Modules Program (SEMP)	–
3.3.11	DI-E-7045	Report, Failure Analysis for the Standard Electronic Modules Program (SEMP)	
3.4	DI-SAFT-80102	Safety Assessment	Tasks 100, 202, and 209 of MIL-STD-882
4.3	DI-T-5315A	First Article Qualification Test Plan	–
4.3	UDI-T-23937	Procedures, Acceptance Test	–
4.4.1.1	DI-NDTI-80566	Test Plan	–
4.4, 4.6.1, and 4.6.2	DI-T-2072	Reports, Test	–
4.6.22	DI-RELI-80249	Environmental Stress Screening (ESS) Report	Task 301 of MIL-STD-785

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and

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Data Requirements List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4.1 Technical manual content. The manual should cover the complete steering system and shall be prepared in accordance with the requirements specified in MIL-M-15071 for type III manuals. The material addressing troubleshooting should direct maintenance personnel to a single starting point when performing fault isolation procedures. Fault isolation should utilize fault logic diagrams based on a fault indication observed during troubleshooting. The diagram should comprise a branching series of questions pertaining to fault isolation. Each question should pertain to a further observation or measurement and result in a "yes" or "no" answer, thereby progressively narrowing the possible functional area of the fault. General type and format should be in accordance with MIL-M-38784.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5.1 Test procedures. The test procedure should state where the tests will be conducted, the sequence in which they will be conducted, test equipment requirements, and detailed step-by-step method of test.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 International interest. Certain provisions of this specification (identified by paragraph number or similar manner, if appropriate) are the subject of international standardization agreement STANAG 4156. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

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6.8 Subject term (key word) listing.

Aft steering unit
Automatic steering
Bridge steering unit
Electrohydraulic steering gear equipment
Local control unit
Steering gear room

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 2030-N029)

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APPENDIX

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included on the drawings when specified in the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWING CONTENTS

30.1 Drawings. The following drawings shall be prepared:

a. Steering control systemproduction drawings
b. Steering system interface control drawings
c. Steering control system installation drawings
d. Steering control system electrical schematic
e. Steering control system wiring diagram.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-17903D(SH)	1991 November 14

3. DOCUMENT TITLE

Steering Control Systems, Naval Shipboard Use

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact(TPOC): Mr. David Breslin (NAVSEA 56W32) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: (703) 602-8015	332-8015
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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